MLDM – Machine Learning and Data Mining

AMD – Aprendizagem e Mineração de Dados

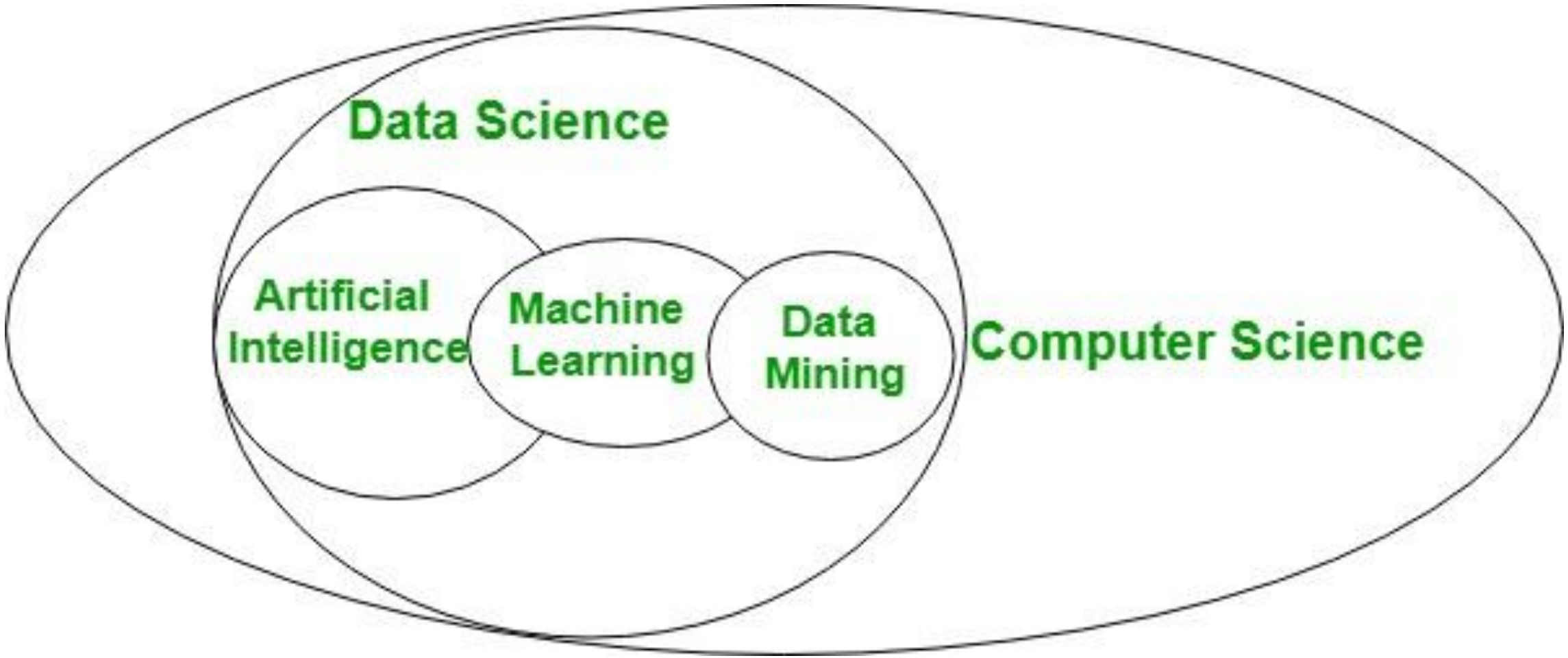
Introduction

(15 September 2026)

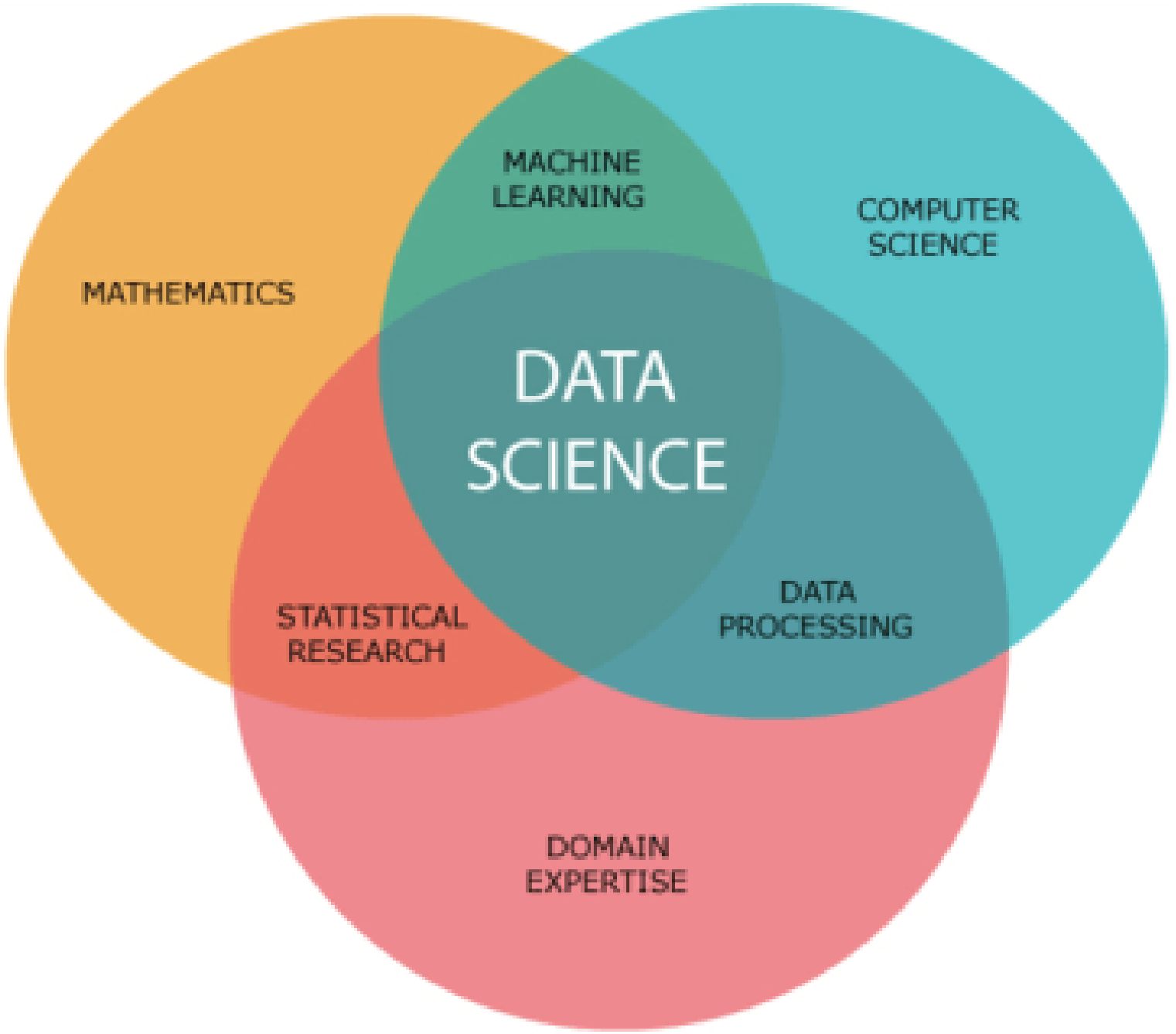
Topics

- Data Science
- Data Mining
- Knowledge Discovery in Databases
- Data, Information, Knowledge, and Wisdom
- Artificial Intelligence
- Machine Learning
  - Supervised, Unsupervised, Semi-supervised, and Reinforcement
- Deep Learning
- Representation and notation

Data Science

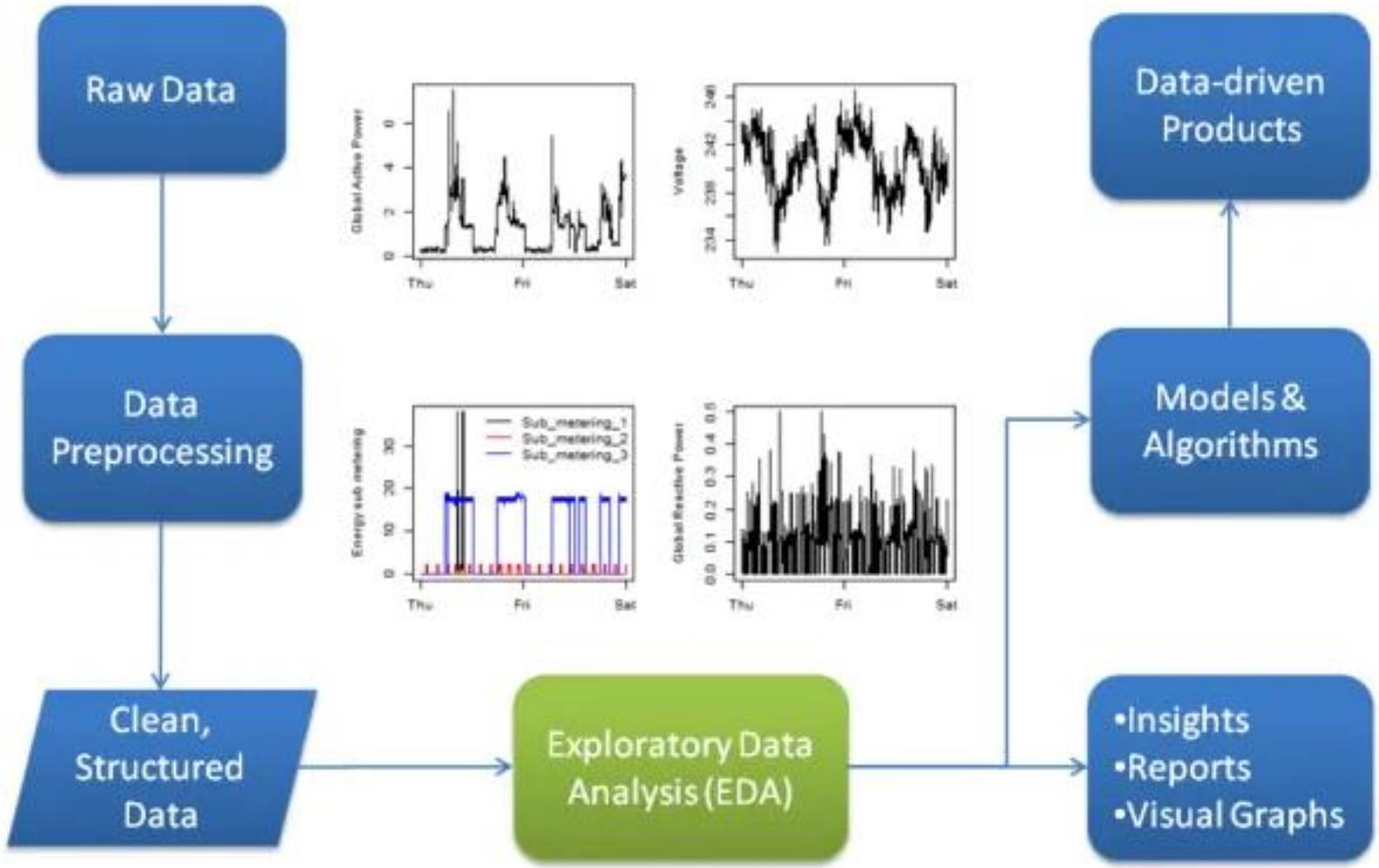


Data Science



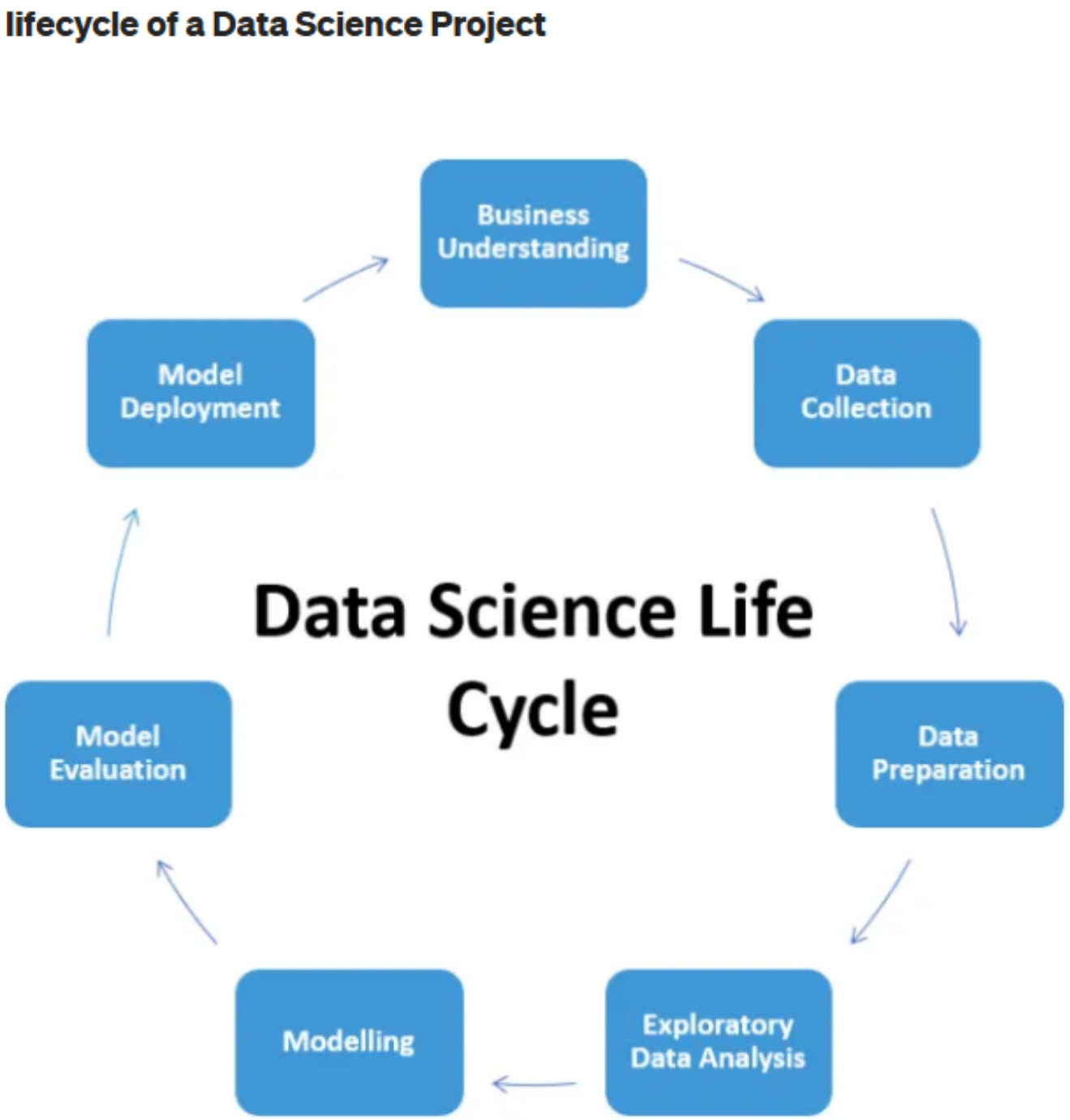
https://medium.com/analytics-vidhya/introduction-to-data-science-28deb32878e7

Data Science



https://medium.com/analytics-vidhya/introduction-to-data-science-28deb32878e7

Data Science



https://medium.com/analytics-vidhya/introduction-to-data-science-28deb32878e7

Data Science

- Data science is an interdisciplinary field
- It sits at the intersection of statistics and computer science
- It employs the scientific method—along with processes, algorithms, and systems—to extract knowledge and make decisions based on data
- The data involved comes in various types and can be structured or unstructured
- It is a field dedicated to the study and organized analysis of data—whether scientific, market-related, financial, social, geographic, historical, biological, psychological, etc.

Data Science

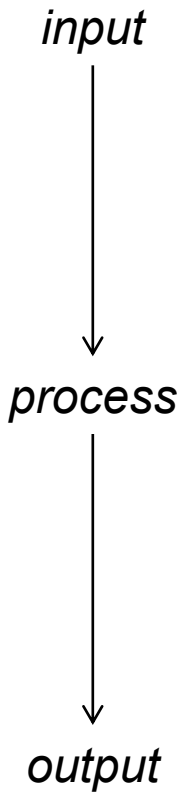
- Knowledge extraction, pattern detection, and/or gaining insights for potential decision-making

- It has existed as a field of knowledge for over 30 years but has gained greater prominence in recent years due to factors such as
  - the emergence and popularization of large databases
  - the development of fields like machine learning

- Data scientists process data to derive business insights, helping companies and organizations make decisions to achieve better results

- Strong connection between the field of data science and AI

From data to...



data within a **context**
(information)

Machine Learning
and Data Mining

data as a **pattern**
(knowledge)

dataset
(examples)

learning/mining
method

model
(learned)

Data Mining

- *Data mining is a practical topic and involves **learning** in a practical, not a theoretical, sense*

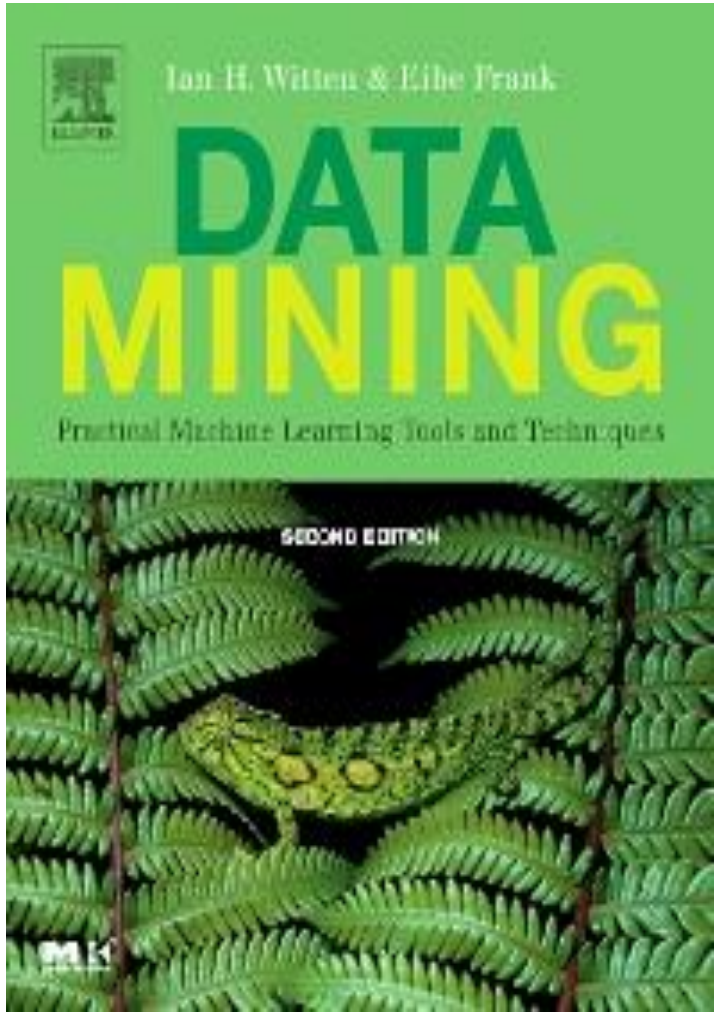
- *We are interested in techniques for finding patterns in data, patterns that provide insight or enable fast and accurate decision making*

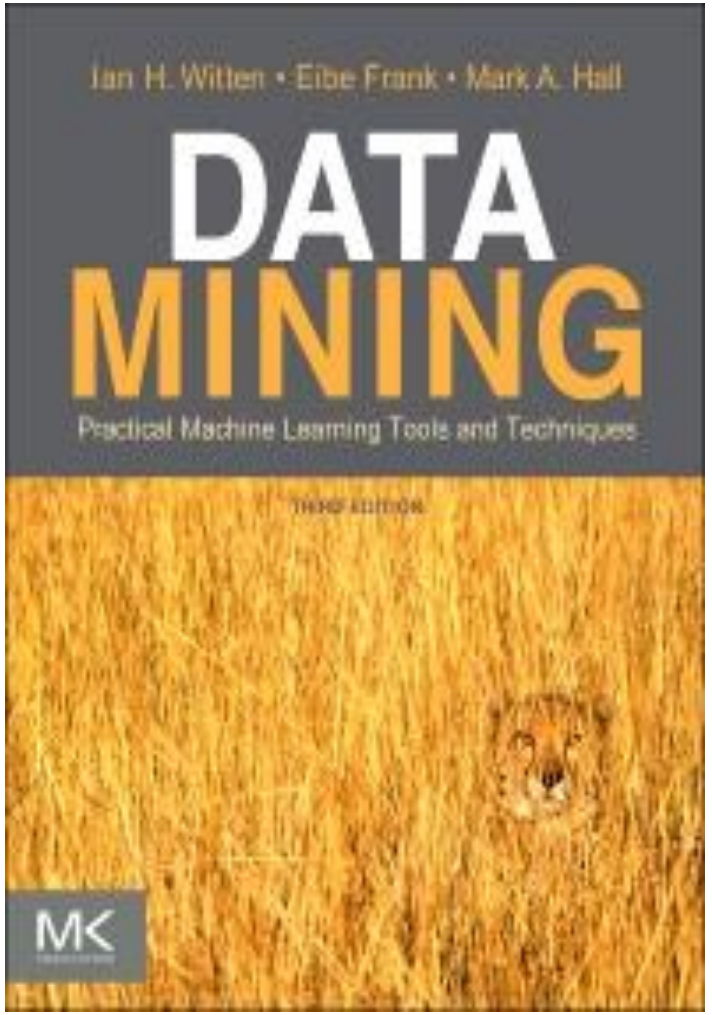


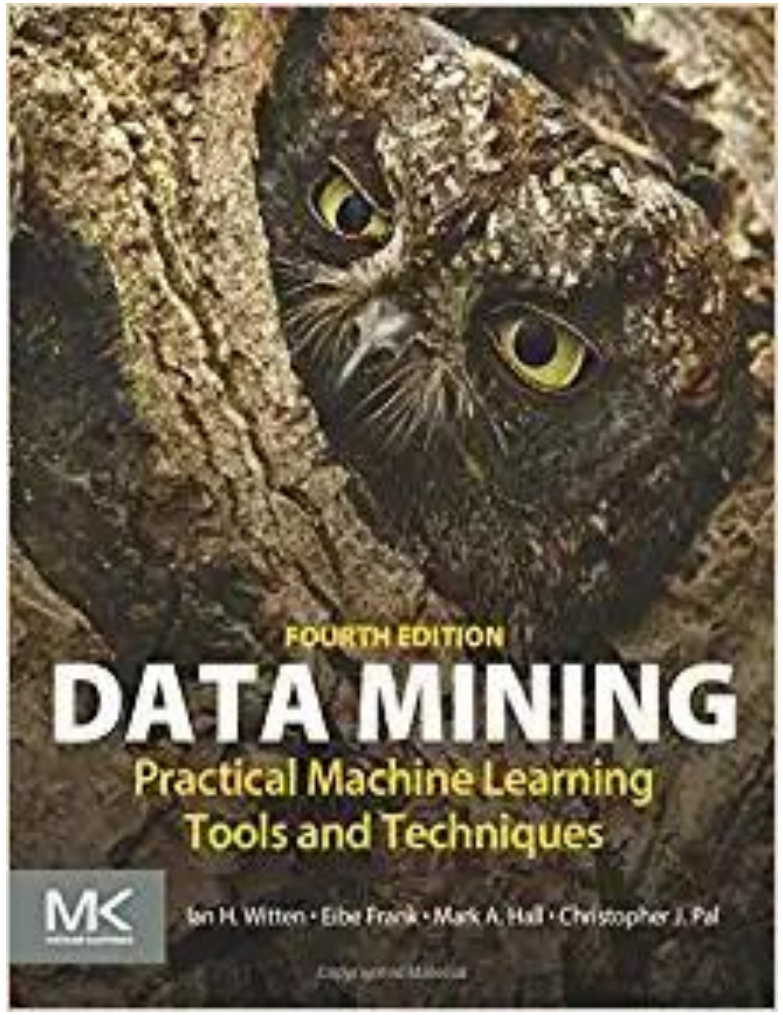


https://en.wikipedia.org/wiki/Gold_mining

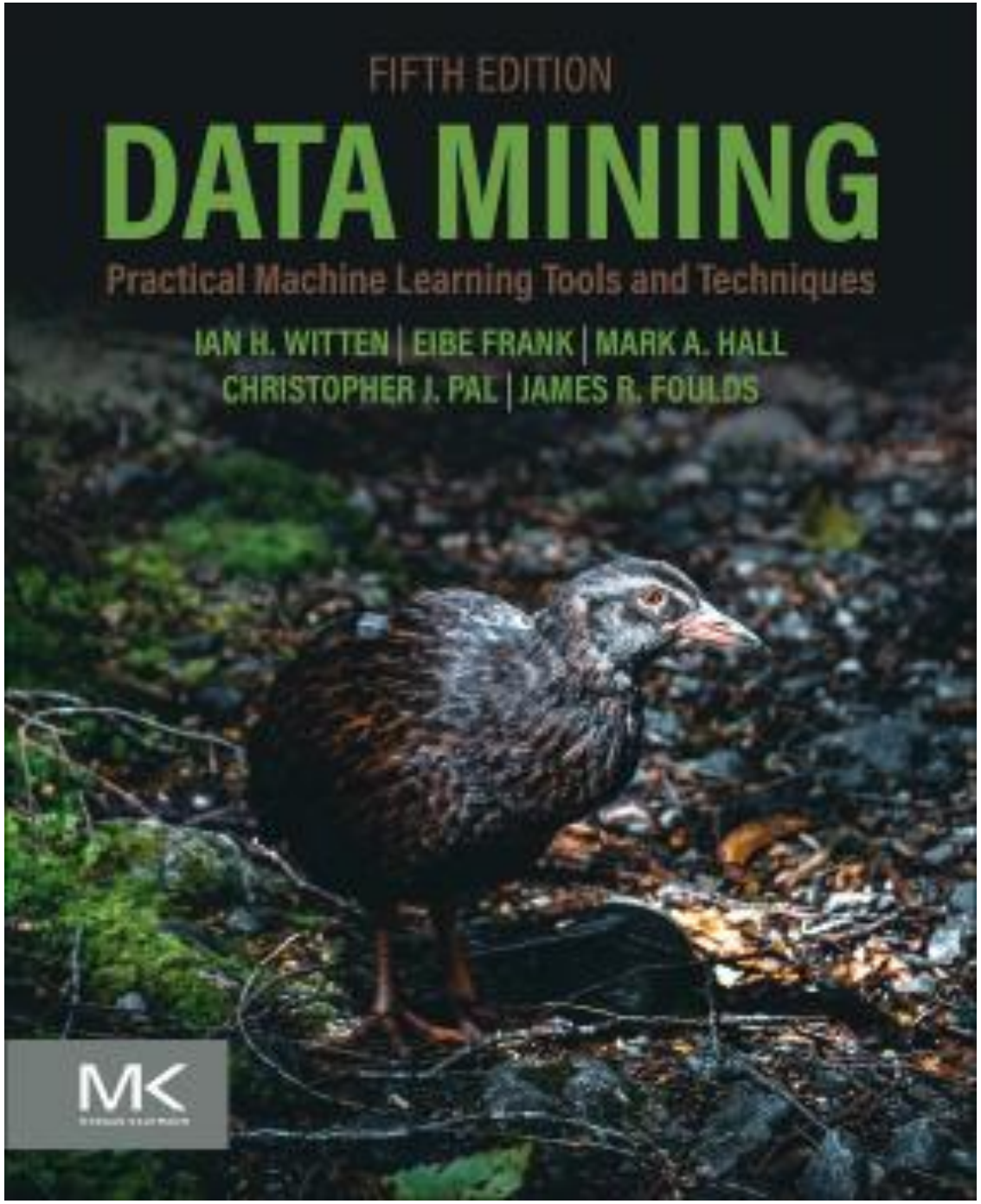
Data Mining


Ian H. Witten & Eibe Frank
DATA
MINING
Practical Machine Learning Tools and Techniques
SECOND EDITION

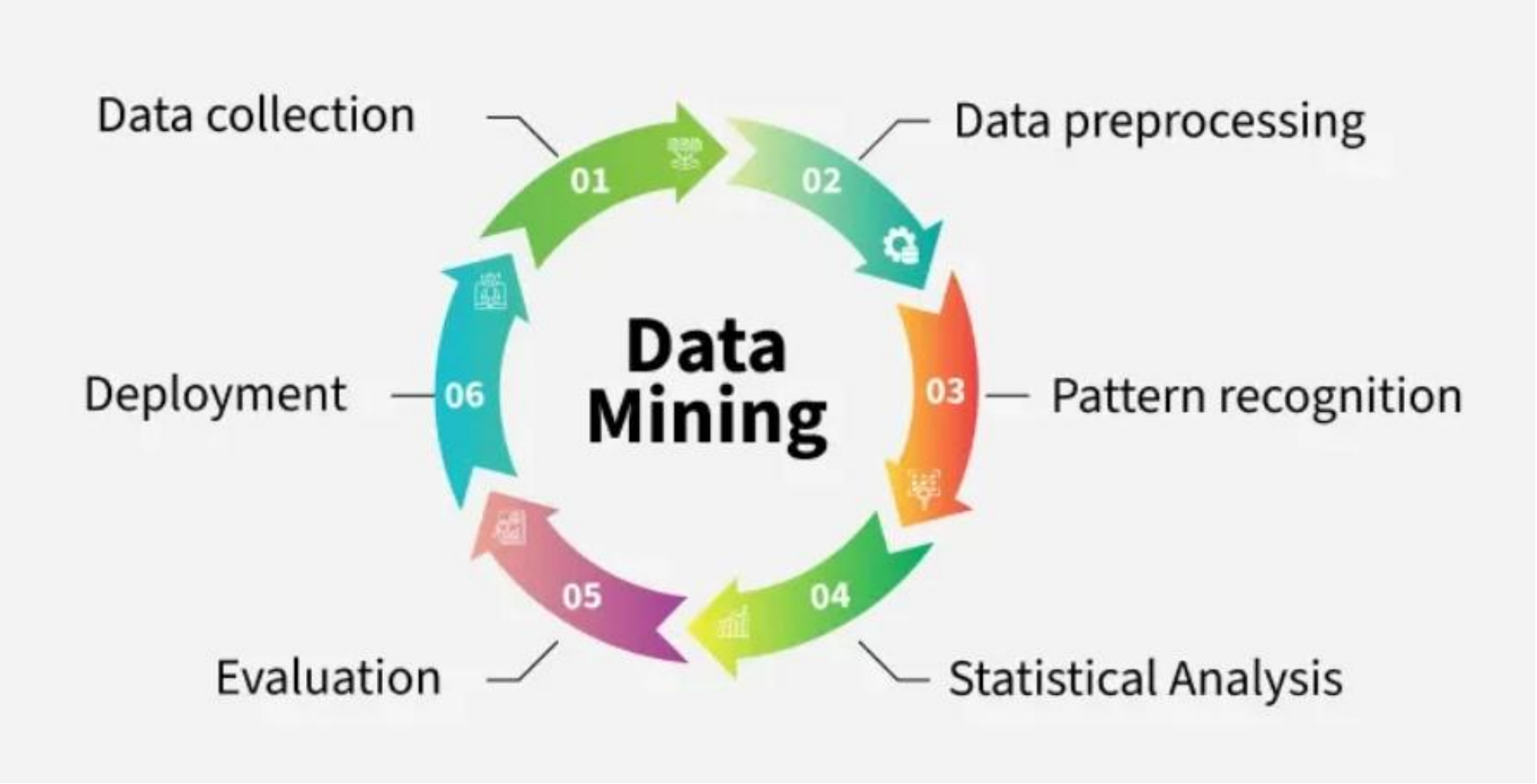

Ian H. Witten • Eibe Frank • Mark A. Hall
DATA
MINING
Practical Machine Learning Tools and Techniques
THIRD EDITION


FOURTH EDITION
DATA MINING
Practical Machine Learning
Tools and Techniques
Ian H. Witten • Eibe Frank • Mark A. Hall • Christopher J. Pal

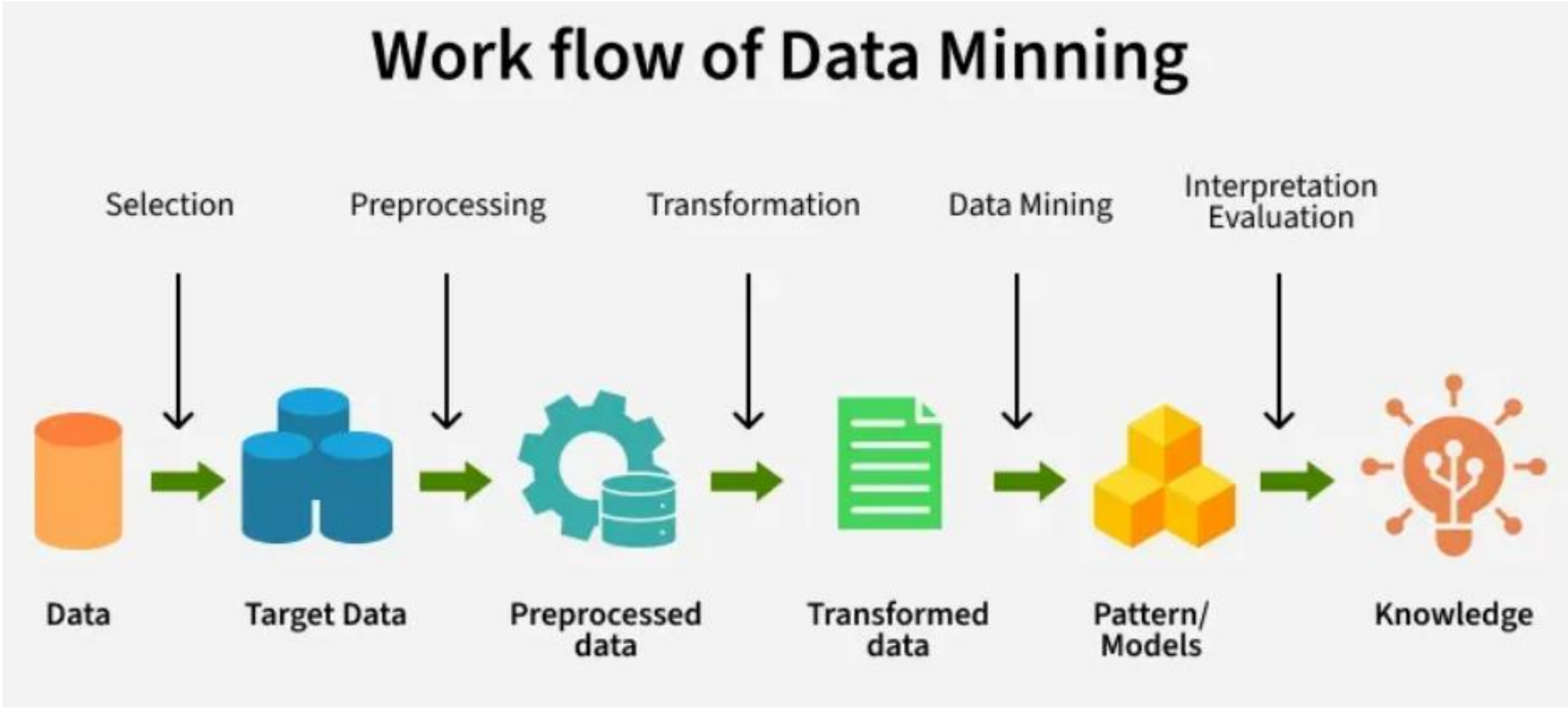
Data Mining


FIFTH EDITION
DATA MINING
Practical Machine Learning Tools and Techniques
IAN H. WITTEN | EIBE FRANK | MARK A. HALL
CHRISTOPHER J. PAL | JAMES R. FOULDS
MK

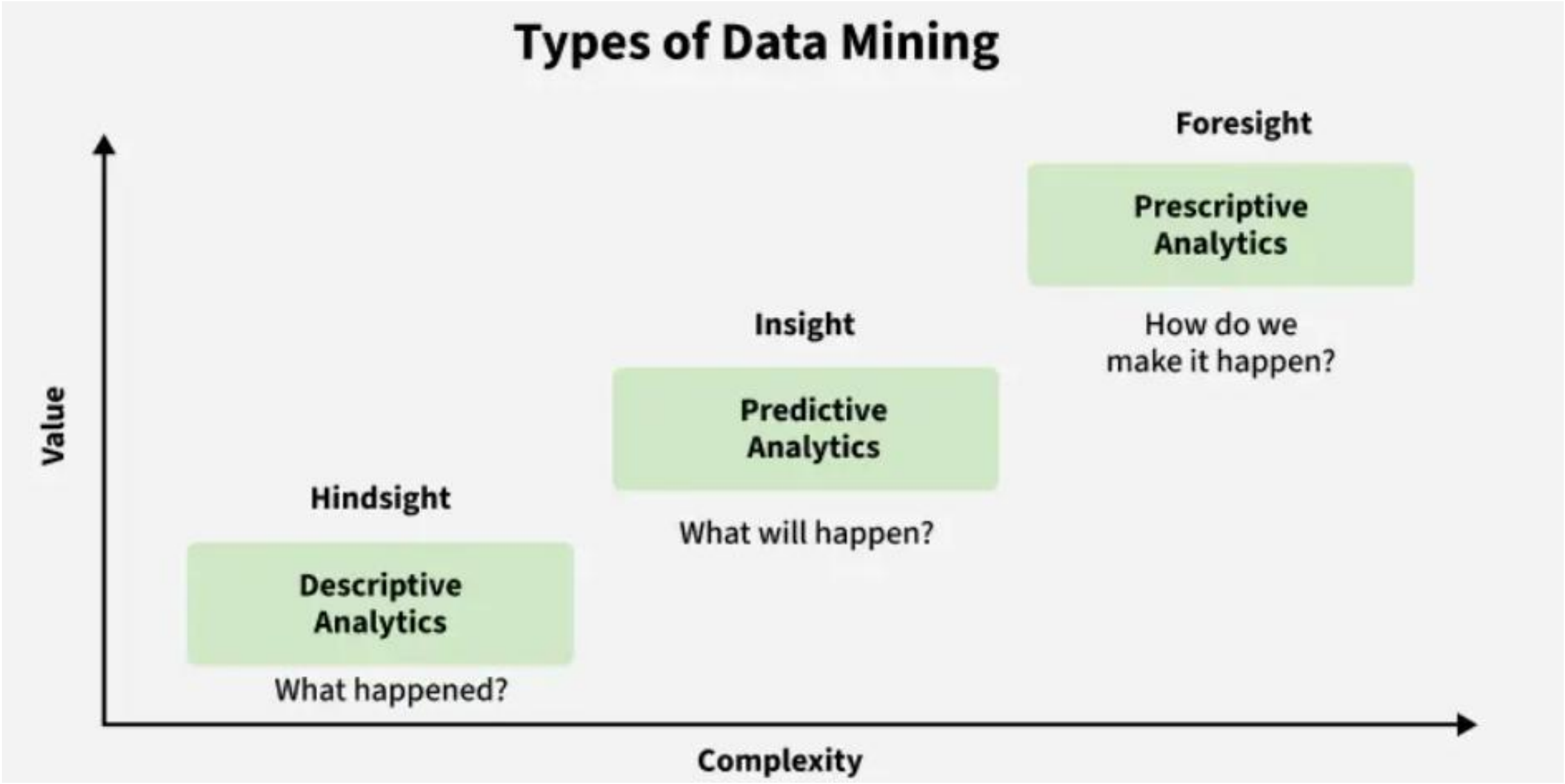
Data Mining



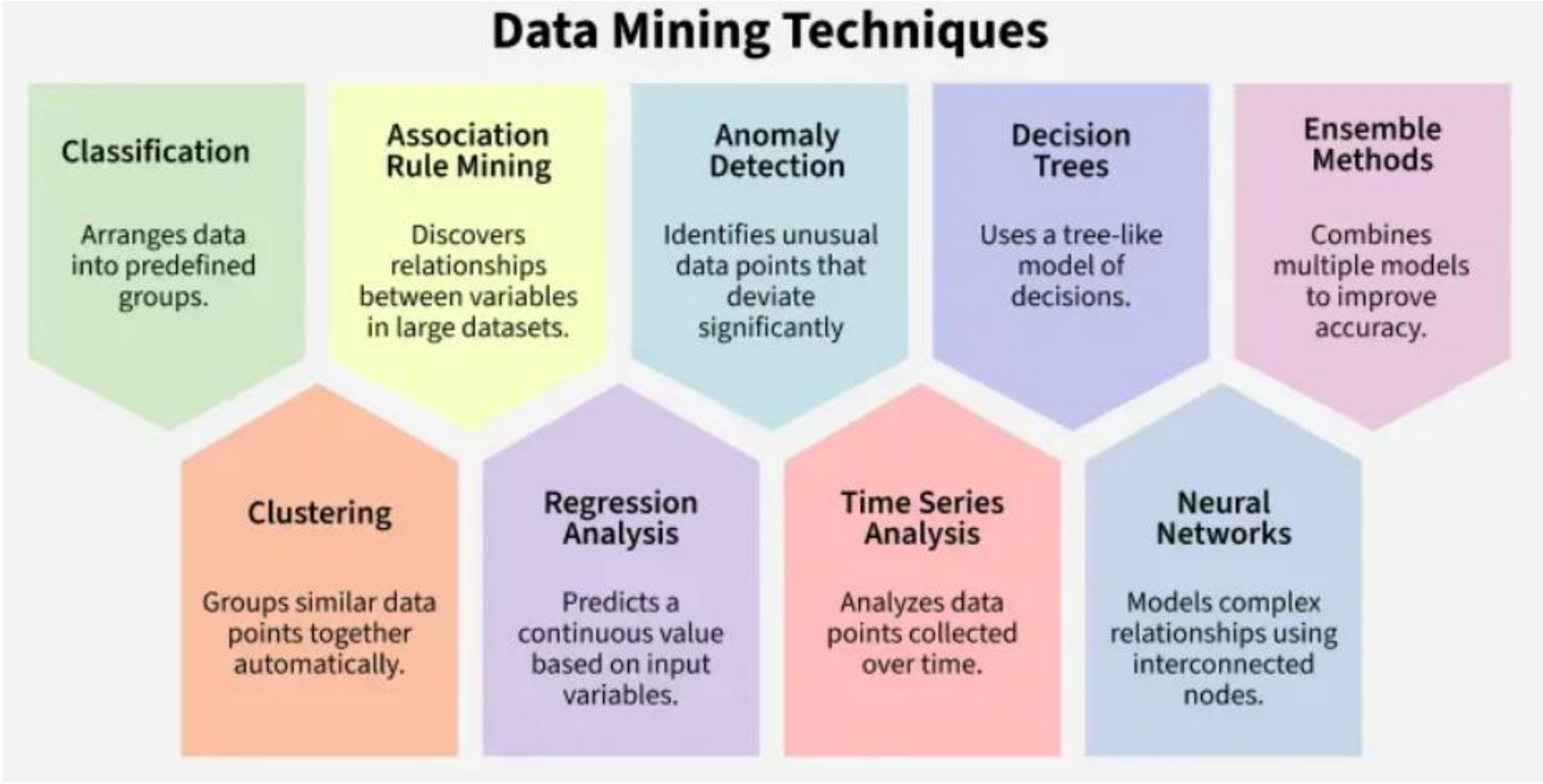
https://www.geeksforgeeks.org/data-science/data-mining/



https://www.geeksforgeeks.org/data-science/data-mining/



https://www.geeksforgeeks.org/data-science/data-mining/



https://www.geeksforgeeks.org/data-science/data-mining/

Data Mining

- **Data mining** is the process of **extracting insights from large datasets** using statistical and computational techniques

- It can involve **structured, semi-structured or unstructured data stored in databases, data warehouses or data lakes**

- The goal is to **uncover hidden patterns** and relationships to **support informed decision-making** and predictions **using methods like clustering, classification, regression, and anomaly detection**

https://www.geeksforgeeks.org/data-science/data-mining/

Data Mining

- Data mining is widely used in industries such as marketing, finance, healthcare and telecommunications

- For example, it helps identify customer segments in marketing or detect disease risk factors in healthcare

- However, it also raises ethical concerns particularly regarding privacy and the misuse of personal data, requiring careful safeguards

https://www.geeksforgeeks.org/data-science/data-mining/

Knowledge Discovery in Databases (KDD)

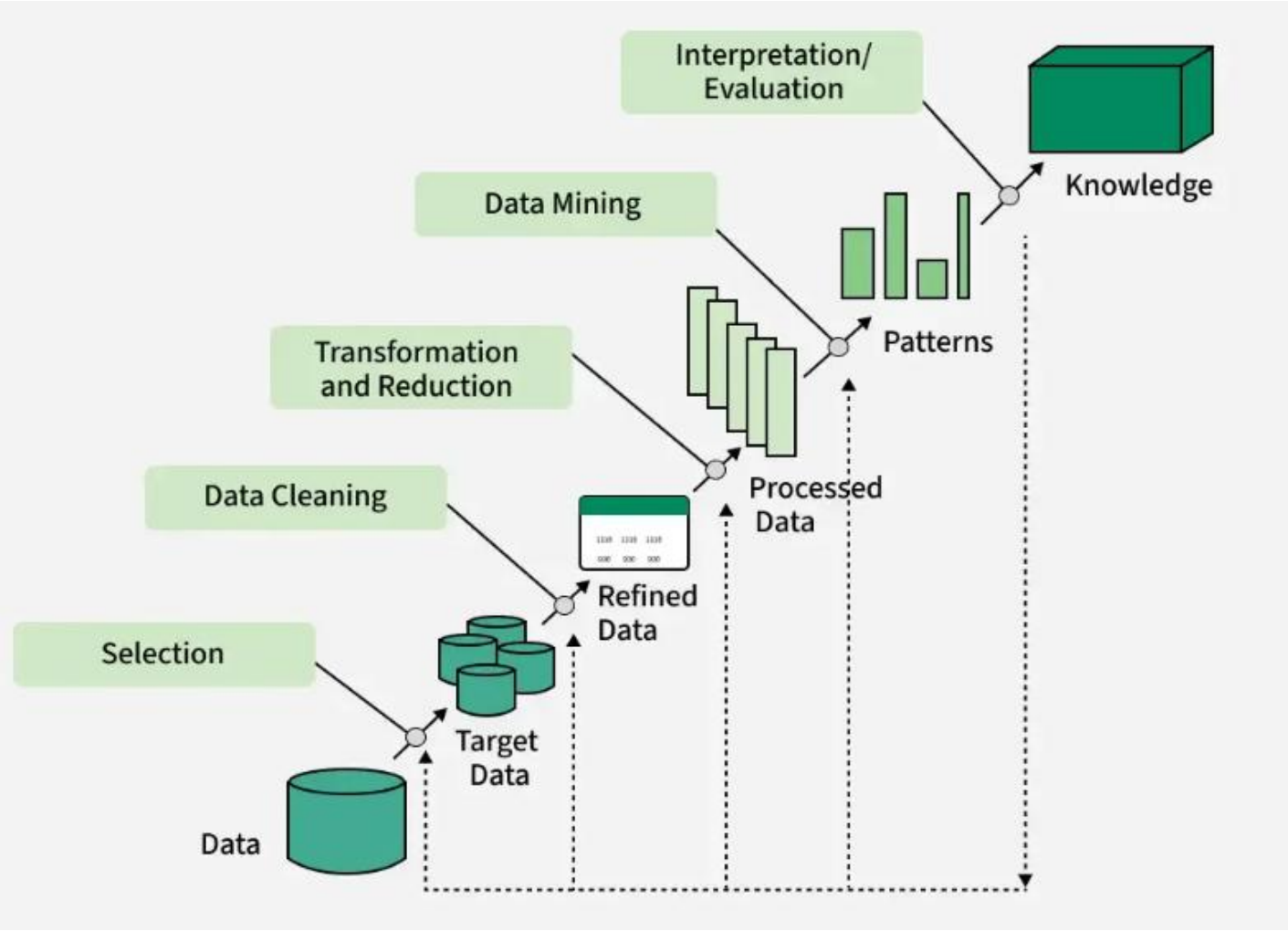
Knowledge Discovery in Databases (KDD) refers to the complete process of **uncovering valuable knowledge** from large datasets

It starts with the **selection of relevant data**,
followed by **preprocessing to clean and organize it,**
transformation to prepare it for analysis,
data mining to uncover patterns and relationships,
and concludes **with the evaluation and interpretation of results,**
ultimately producing valuable knowledge or insights

KDD is widely utilized in fields like machine learning, pattern recognition, statistics, artificial intelligence, and data visualization

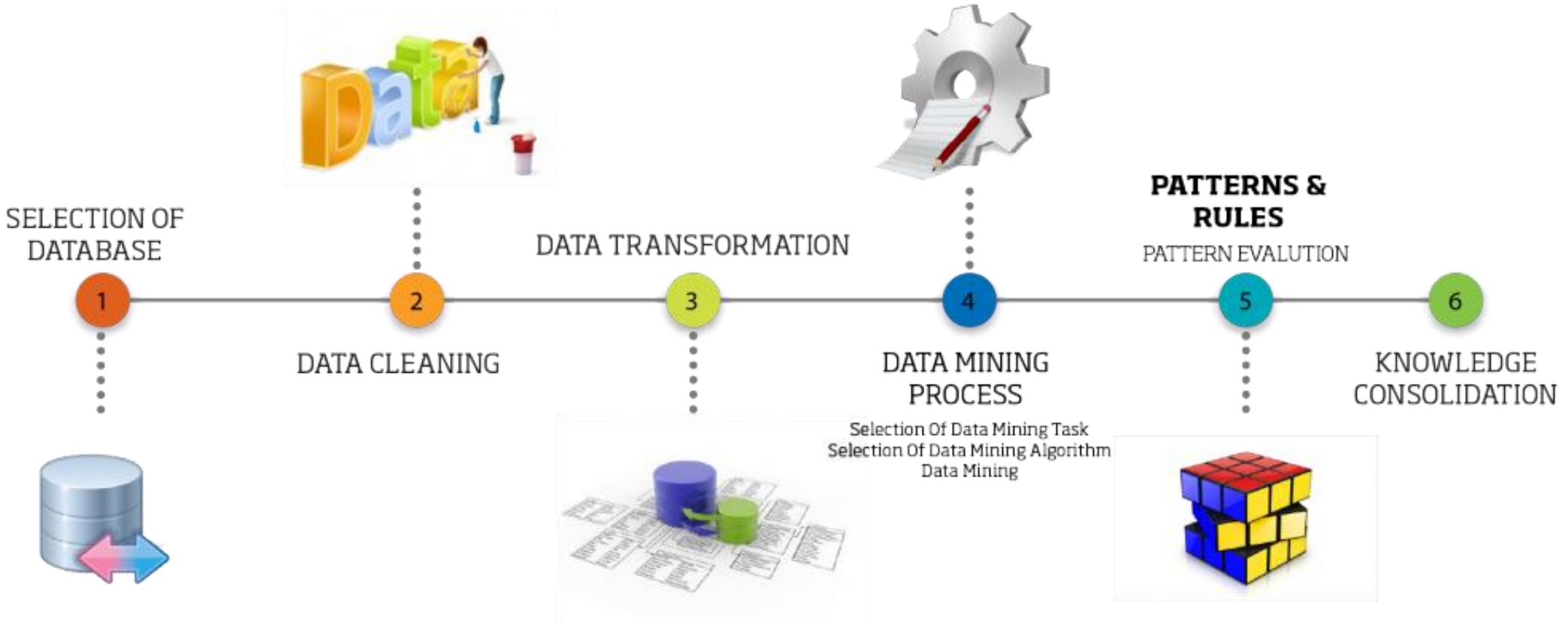
https://www.geeksforgeeks.org/dbms/kdd-process-in-data-mining/

Knowledge Discovery in Databases (KDD)



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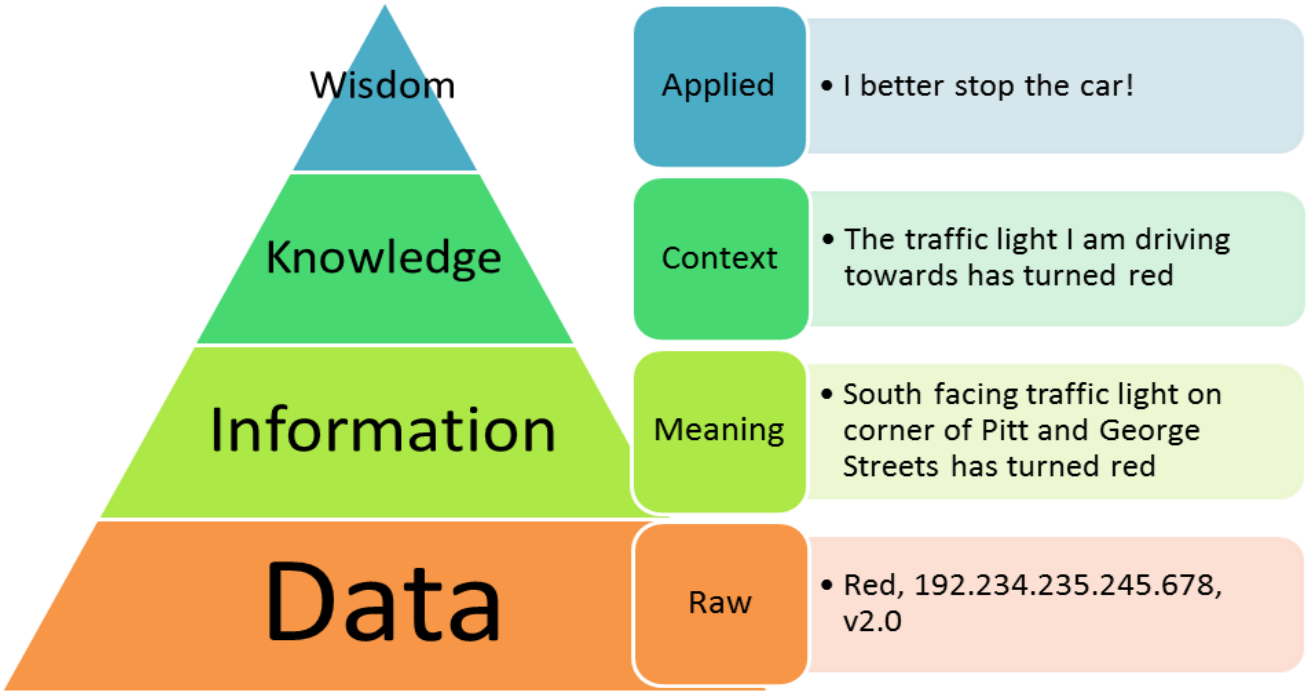
Knowledge Discovery in Databases (KDD)

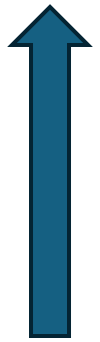


https://www.cognixia.com/blog/data-mining-integral-part-knowledge-discovery-process/

DIKW

The Data, Information, Knowledge, and Wisdom (DIKW) hierarchy

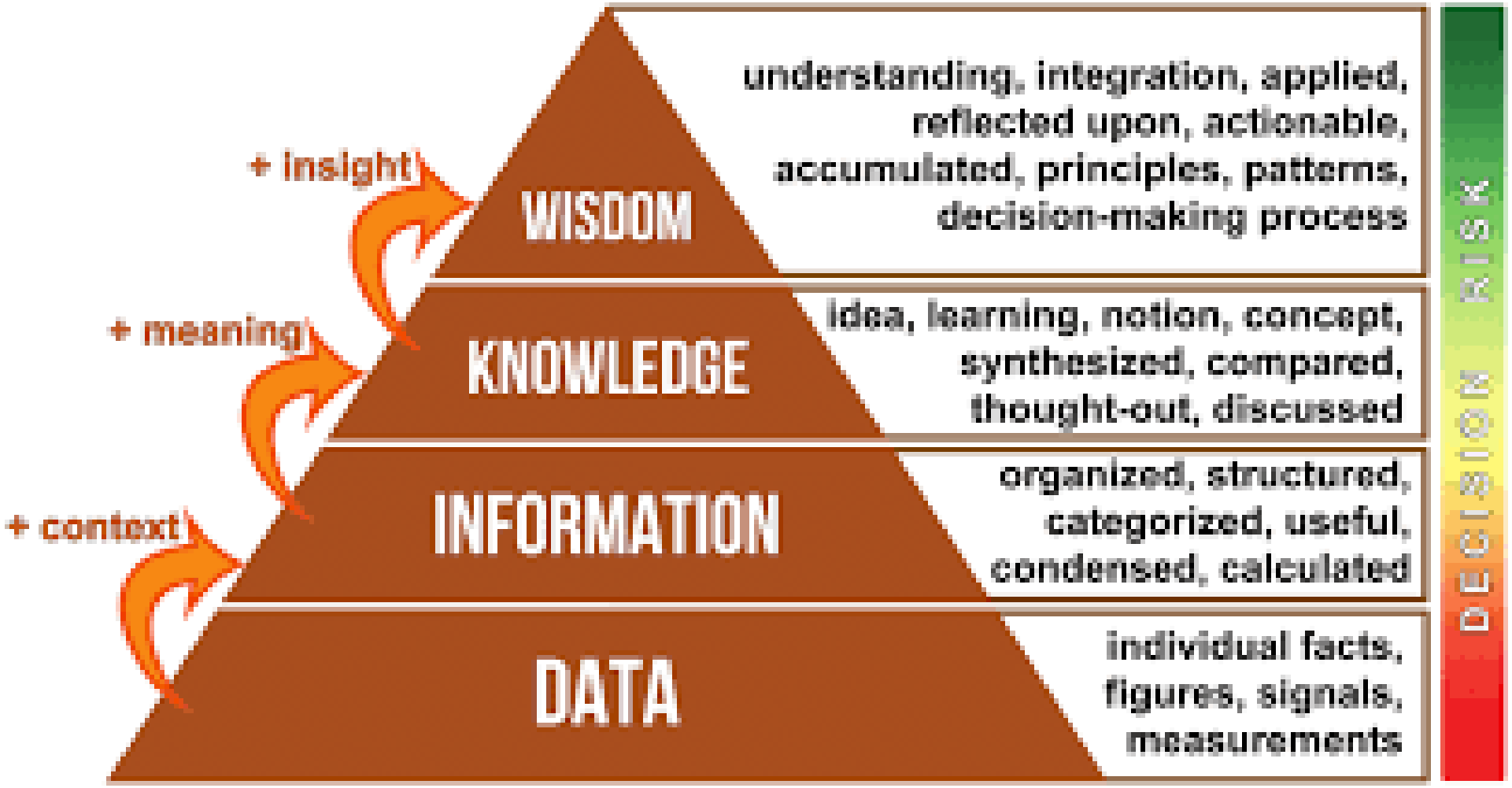




Machine Learning and Data Mining tasks

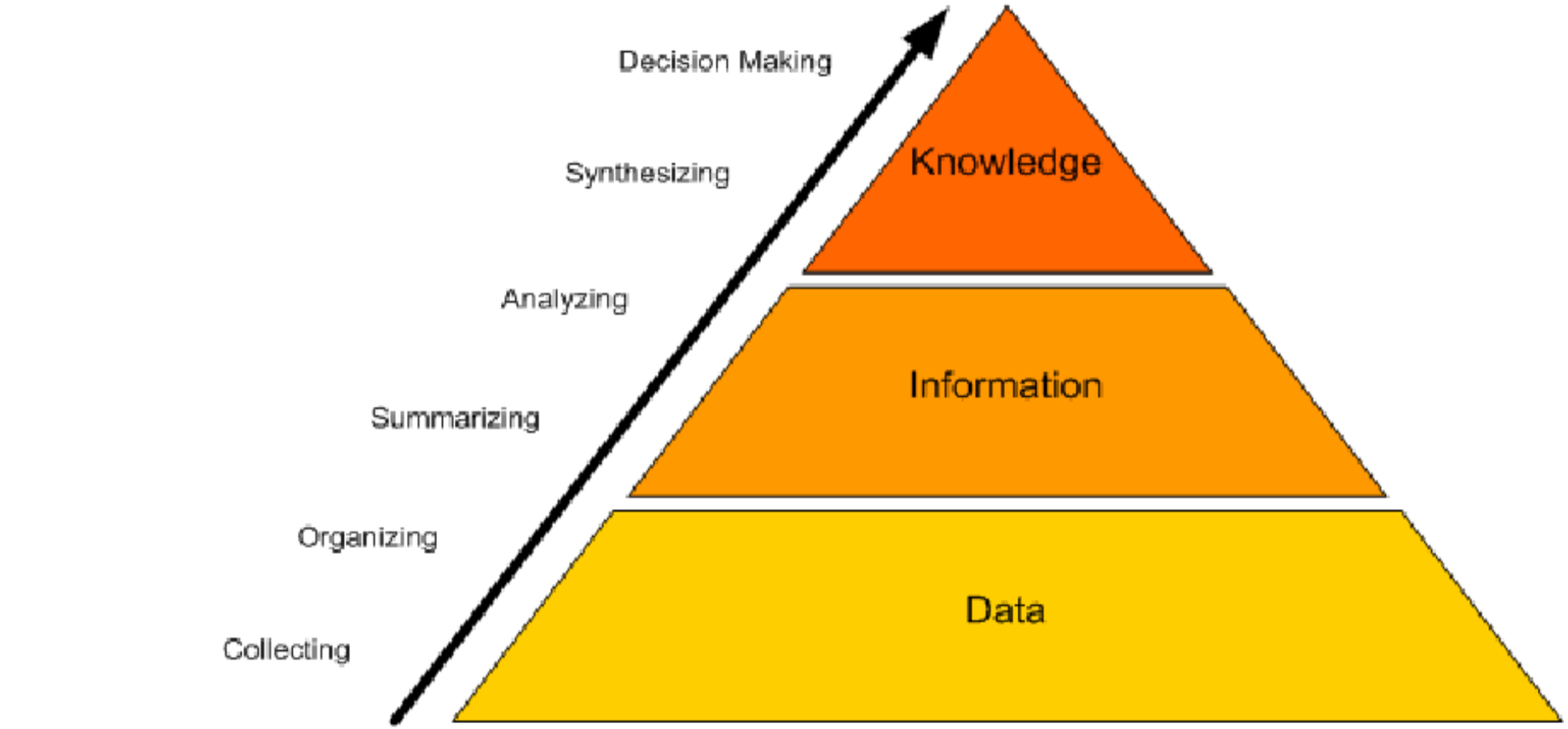
https://allthingy.com/data-information-knowledge-wisdom/

DIKW



https://www.researchgate.net/figure/The-data-information-knowledge-wisdom-DIKW-hierarchy-as-a-pyramid-to-manage-knowledge_fig6_332400827

DIK



Finck, 2005)

Figure 1: DIK Model

https://www.semanticscholar.org/paper/Revisiting-Data-%2C-Information-%2C-Knowledge-and-(-)-Mutongi/14b904f0e28b26eaa214fe086f0df90ce502ae53

DIKW Model - Taxonomy

- **Data**

Symbols and representations e.g., 2010.OCT.5, 155 euros, Miguel, 12-CD-45

- **Information**

Processed data that is useful answers the four "W" questions (4Ws) who, what, where, when

- **Knowledge**

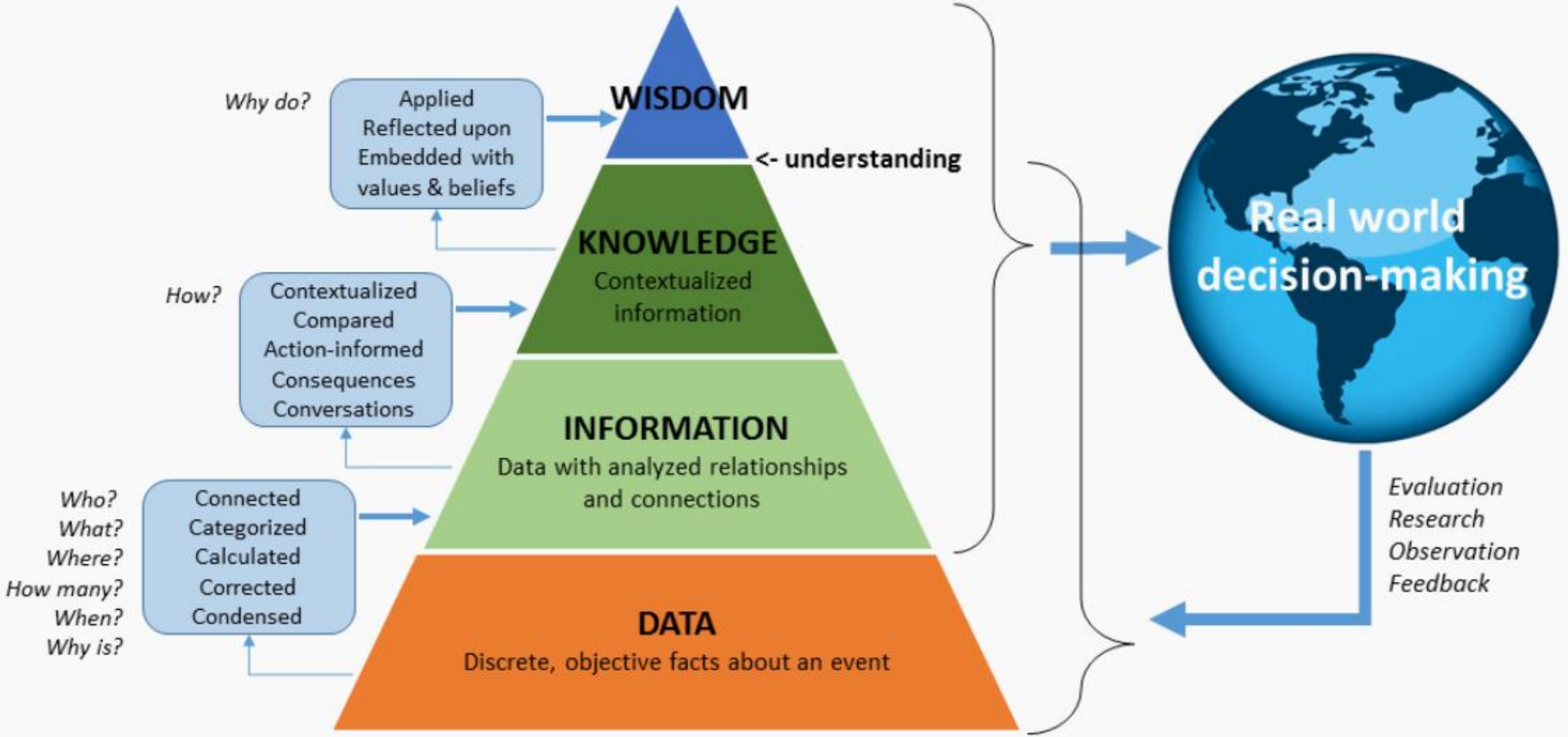
The use of data and information answers questions regarding "how"

understanding of the "how“

- **Wisdom**

Evaluation of the understanding gained

Modelo DIKW – Decision

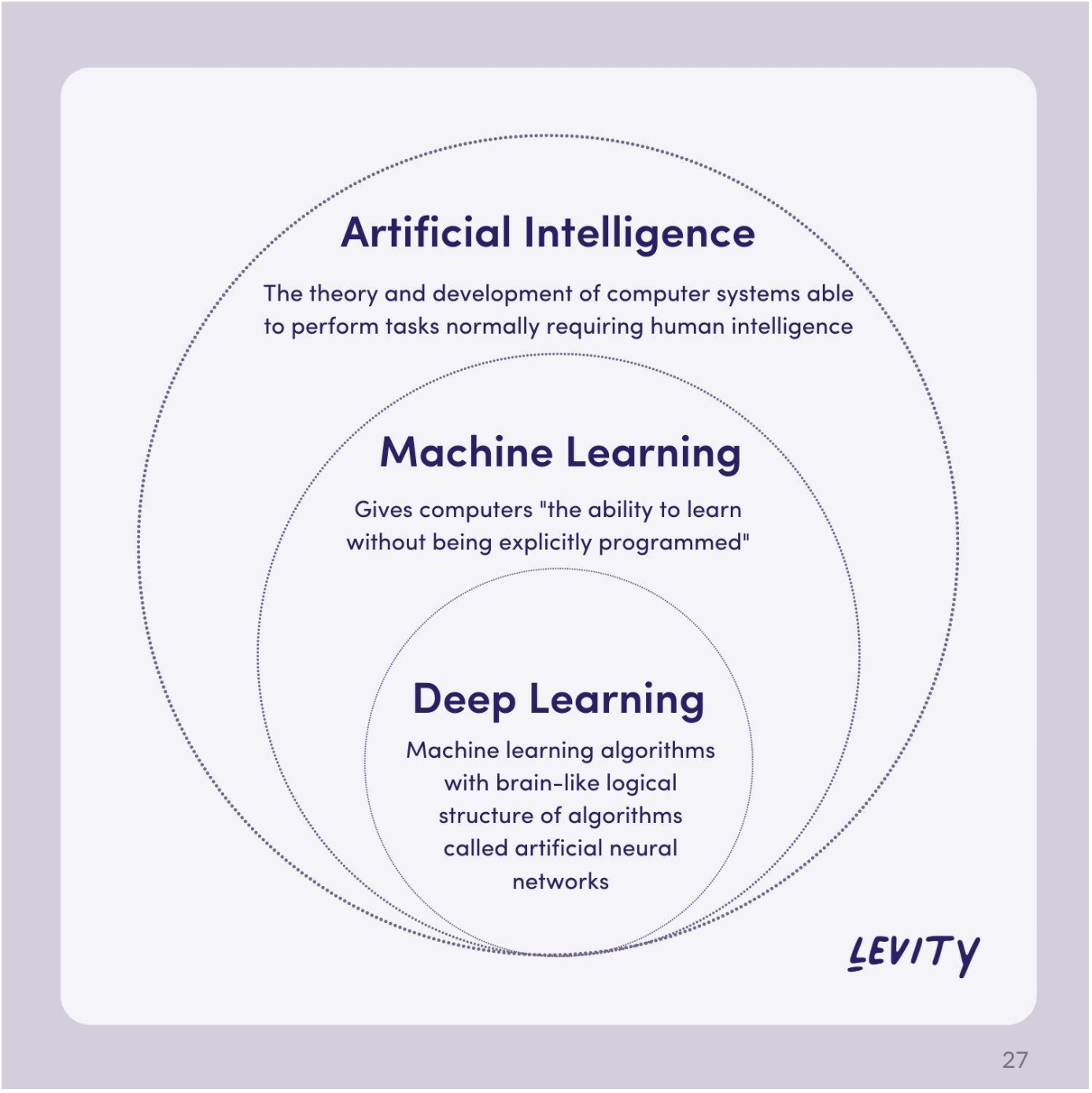


https://universidadedatecnologia.com.br/dados-vs-informacoes-digitais/data-information-knowledge-wisdom/

Artificial Intelligence

Machine Learning

Deep Learning

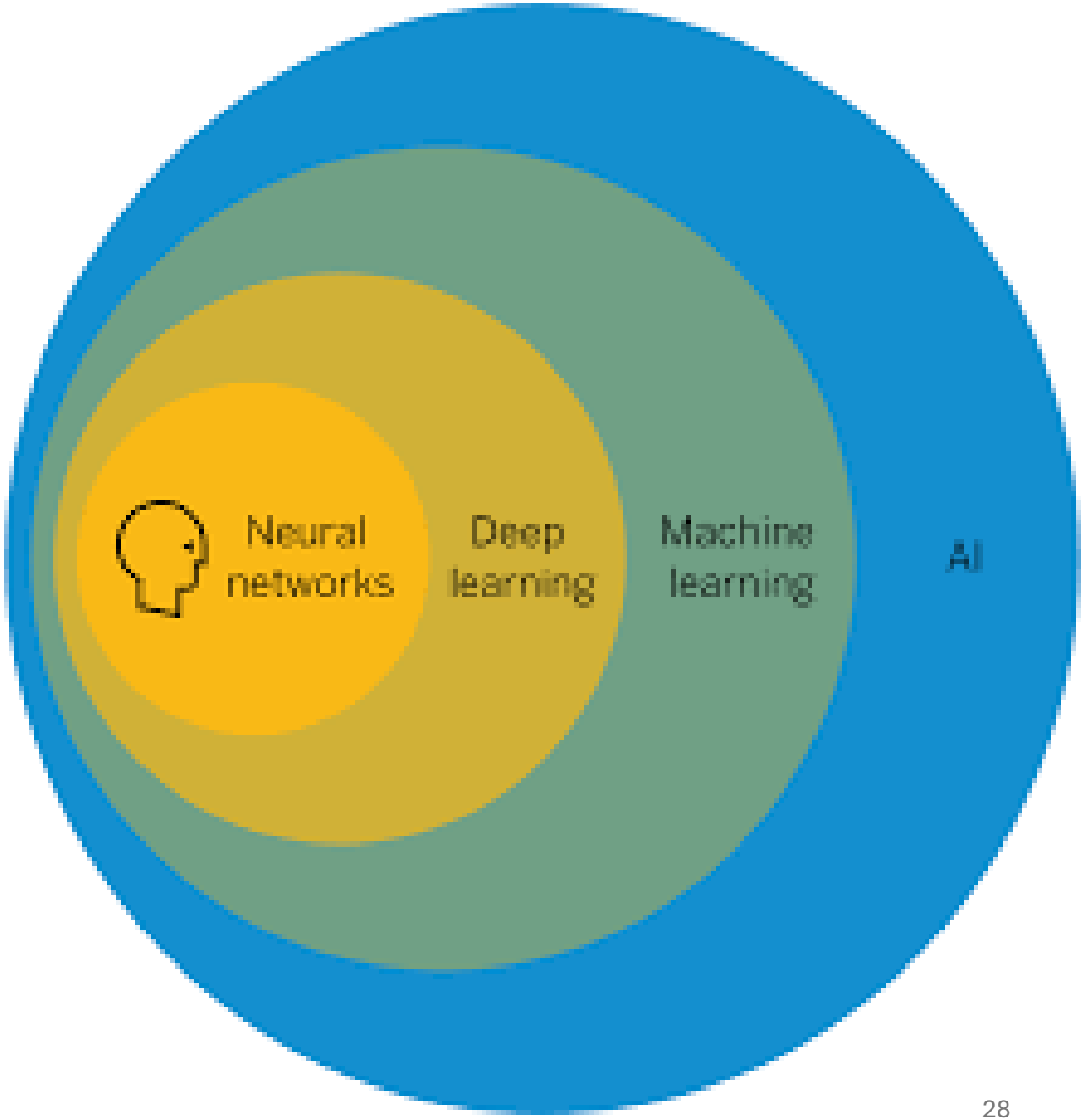


Artificial Intelligence

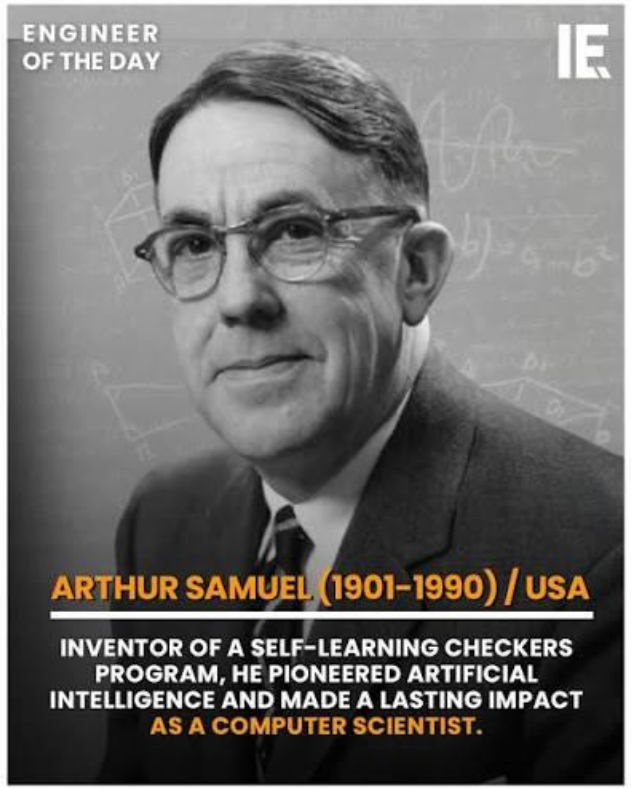
Machine Learning

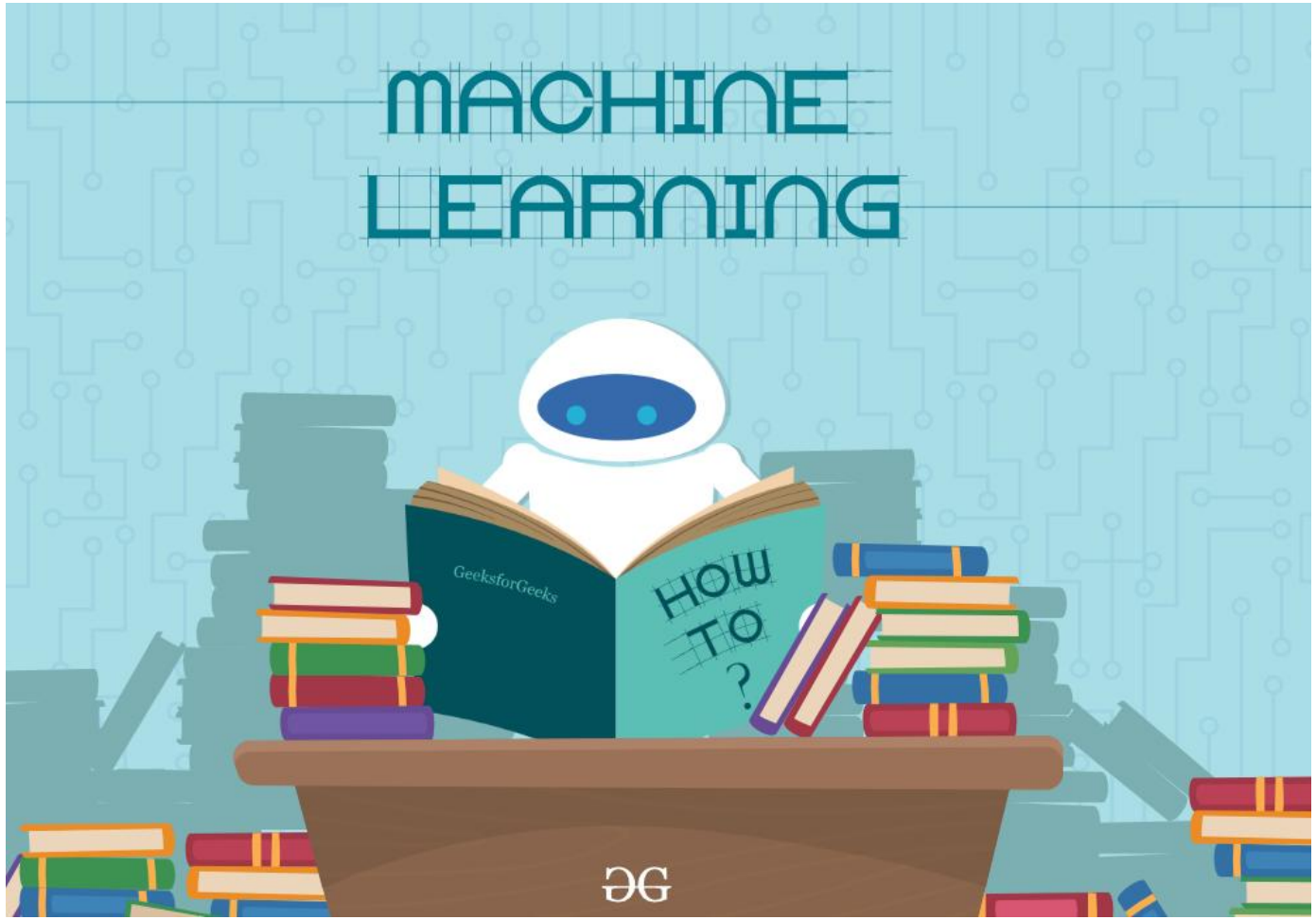
Deep Learning

Neural Networks



Aprendizagem Automática (*Machine Learning*)

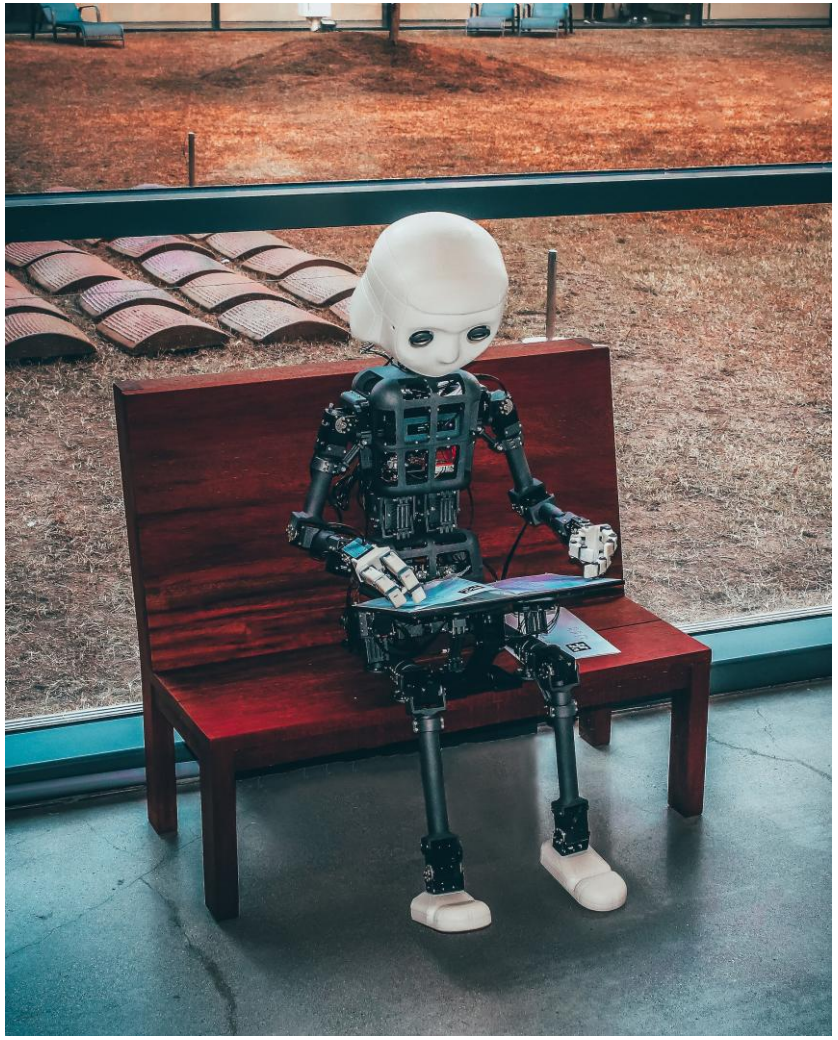




Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed

(Arthur Samuel)

Machine Learning



Machine learning is the study of computer algorithms that allow computer programs to automatically improve through experience

(Thomas Mitchell)

Machine Learning



A machine can learn? How?

How humans learn?

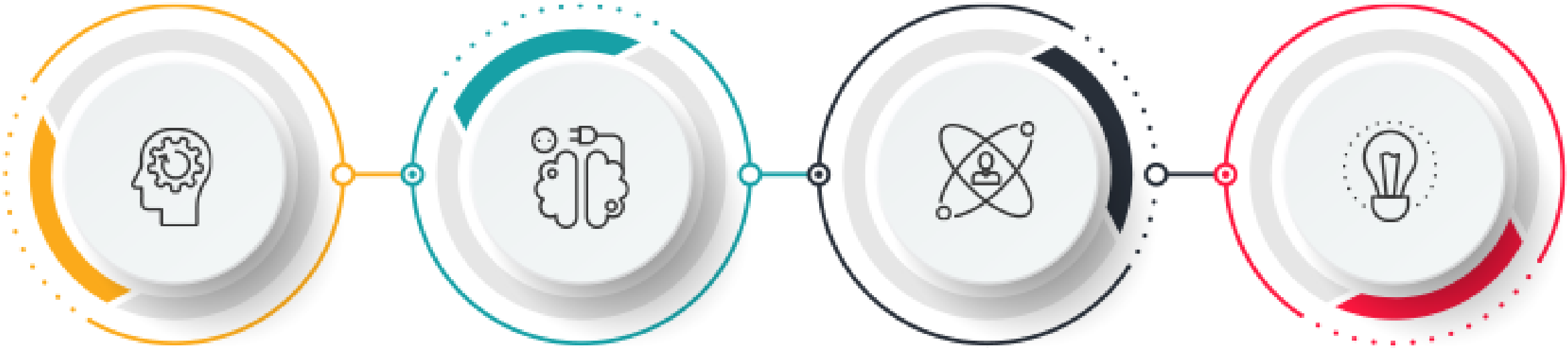
Machine learning is about:
automating automation
making computers program themselves

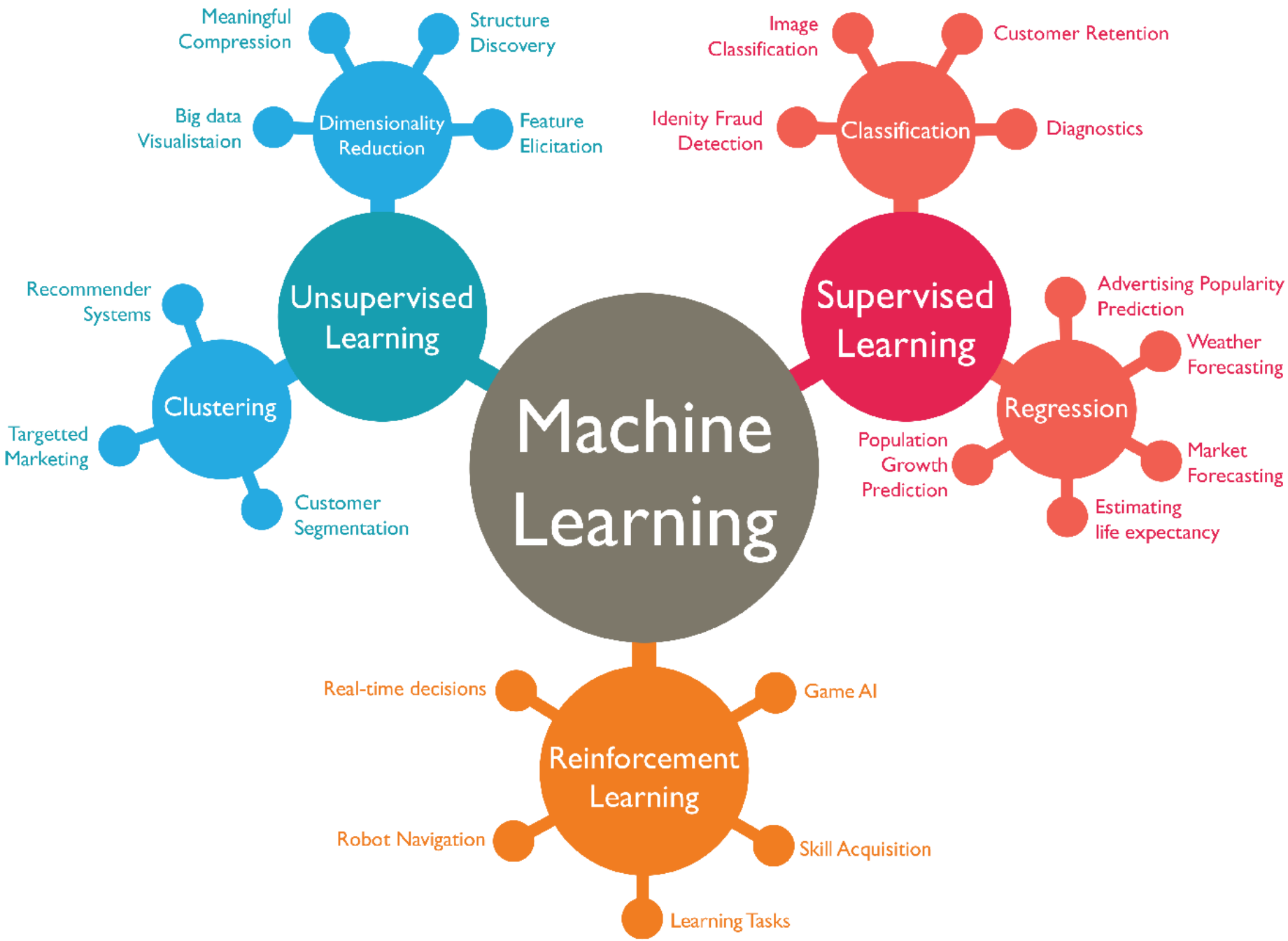
A computer program is said to learn from experience E with respect to some set of tasks T and performance measure P

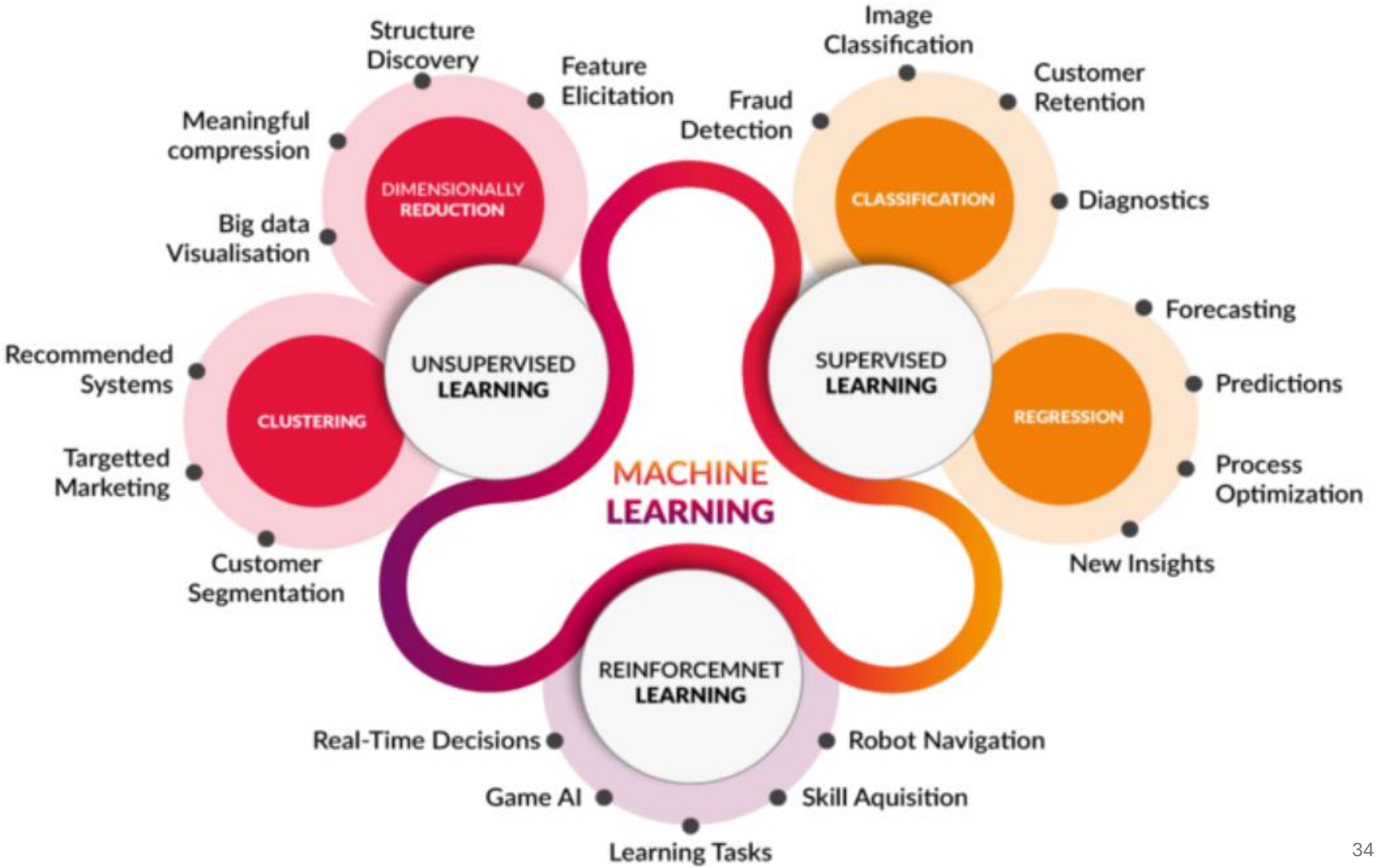
if

its performance at tasks in T, as measured by P, improves with experience E.

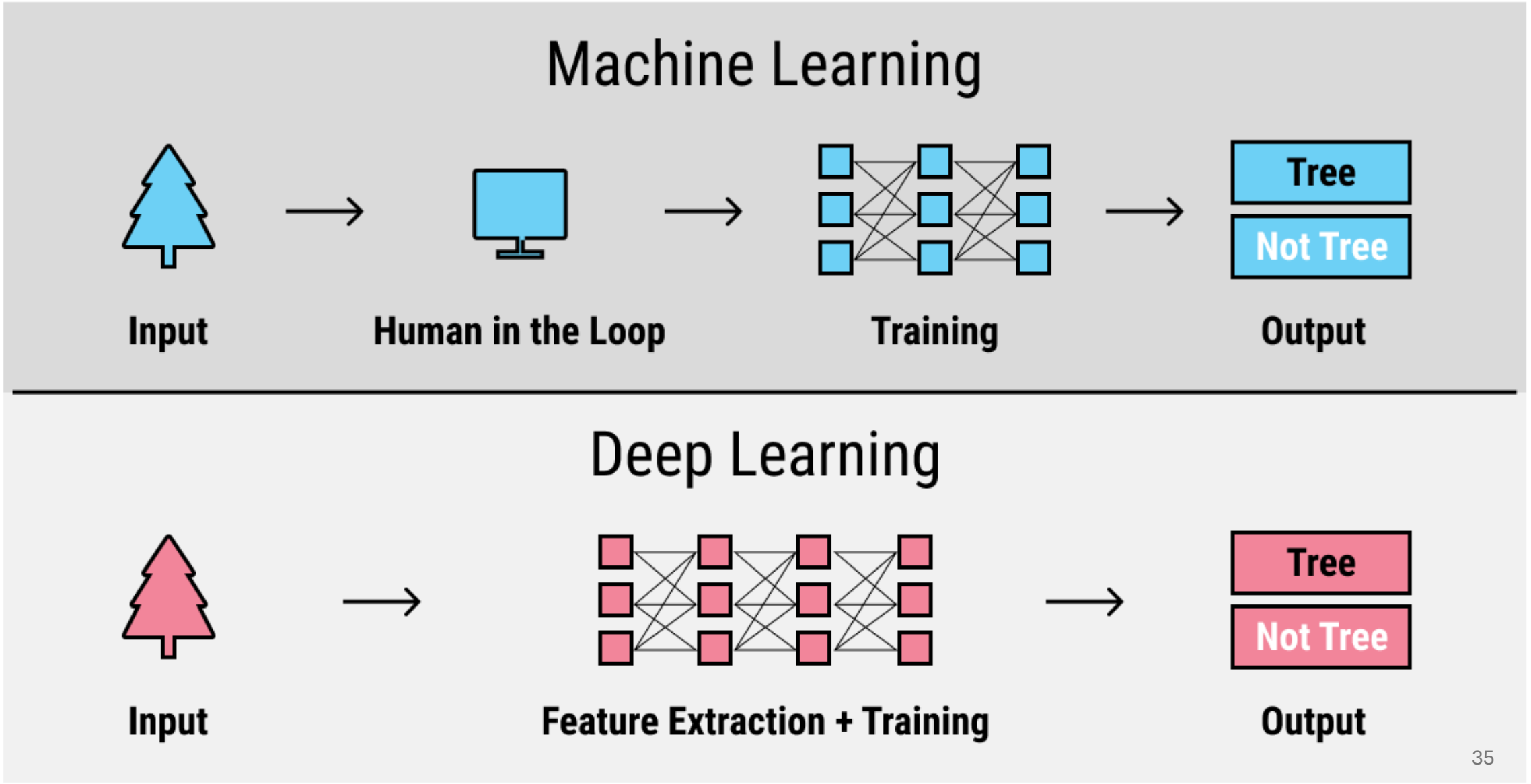
Types of Machine Learning




Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

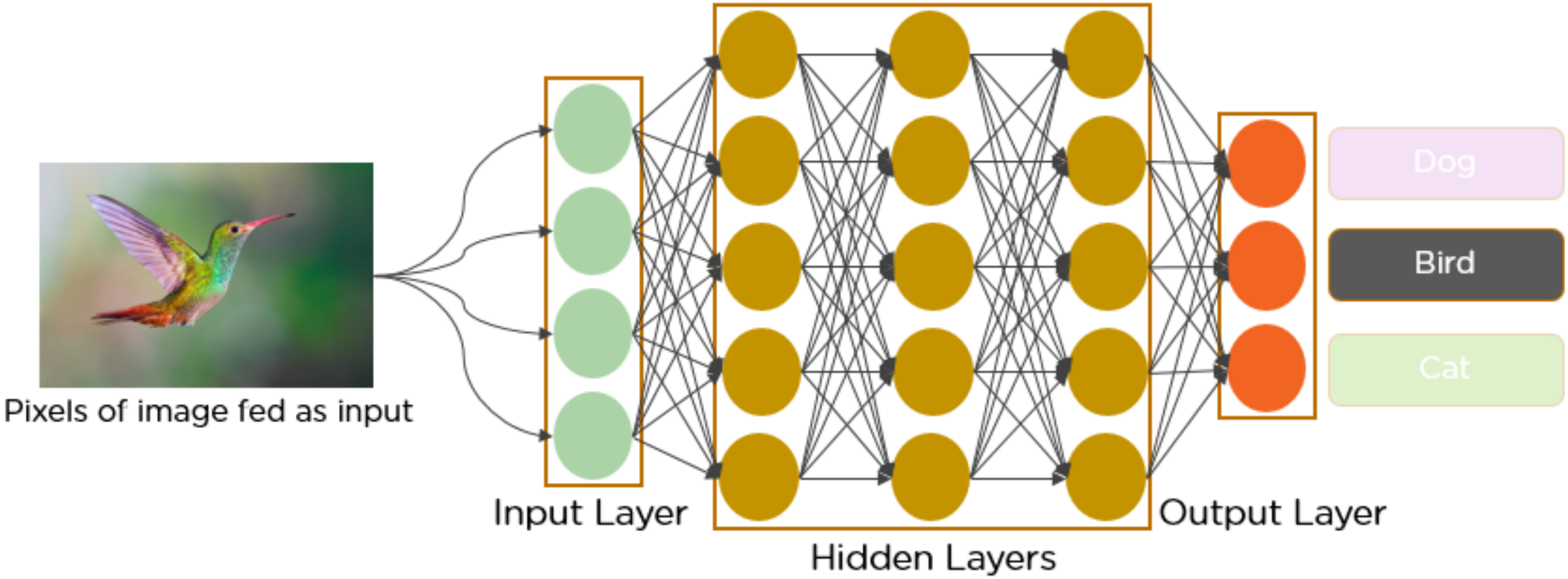

Structure Discovery
Feature Elicitation
Meaningful compression
Big data Visualisation
DIMENSIONALLY REDUCTION
Image Classification
Customer Retention
Fraud Detection
CLASSIFICATION
Diagnostics
Recommended Systems
Targetted Marketing
CLUSTERING
Customer Segmentation
UNSUPERVISED LEARNING
SUPERVISED LEARNING
MACHINE LEARNING
Forecasting
Predictions
Process Optimization
REGRESSION
New Insights
REINFORCEMNET LEARNING
Real-Time Decisions
Robot Navigation
Game AI
Skill Aquisition
Learning Tasks

Machine Learning and Deep Learning



Machine Learning and Deep Learning

Use of Neural Networks



Data, Database, and Dataset



Database

Export
(View)

age	prescription	astigmatic	tear_rate	lenses
young	myope	yes	normal	hard
young	myope	no	normal	soft
young	hypermetrope	yes	reduced	none
young	hypermetrope	no	normal	soft
young	hypermetrope	no	reduced	none
presbyopic	myope	yes	reduced	none
presbyopic	myope	yes	normal	hard
presbyopic	hypermetrope	yes	reduced	none
presbyopic	hypermetrope	yes	normal	none
presbyopic	hypermetrope	no	normal	soft
presbyopic	hypermetrope	no	reduced	none
pre-presbyopic	myope	yes	reduced	none
pre-presbyopic	myope	yes	normal	hard
pre-presbyopic	myope	no	normal	soft
pre-presbyopic	hypermetrope	yes	normal	none
pre-presbyopic	hypermetrope	no	normal	soft

Dataset

Instances (rows)
and
Attributes (columns)

Data and tasks

age	prescription	astigmatic	tear_rate	lenses
young	myope	yes	normal	hard
young	myope	no	normal	soft
young	hypermetrope	yes	reduced	none
young	hypermetrope	no	normal	soft
young	hypermetrope	no	reduced	none
presbyopic	myope	yes	reduced	none
presbyopic	myope	yes	normal	hard
presbyopic	hypermetrope	yes	reduced	none
presbyopic	hypermetrope	yes	normal	none
presbyopic	hypermetrope	no	normal	soft
presbyopic	hypermetrope	no	reduced	none
pre-presbyopic	myope	yes	reduced	none
pre-presbyopic	myope	yes	normal	hard
pre-presbyopic	myope	no	normal	soft
pre-presbyopic	hypermetrope	yes	normal	none
pre-presbyopic	hypermetrope	no	normal	soft

Dataset



Identify/label concepts

Classification



Estimate/predict values

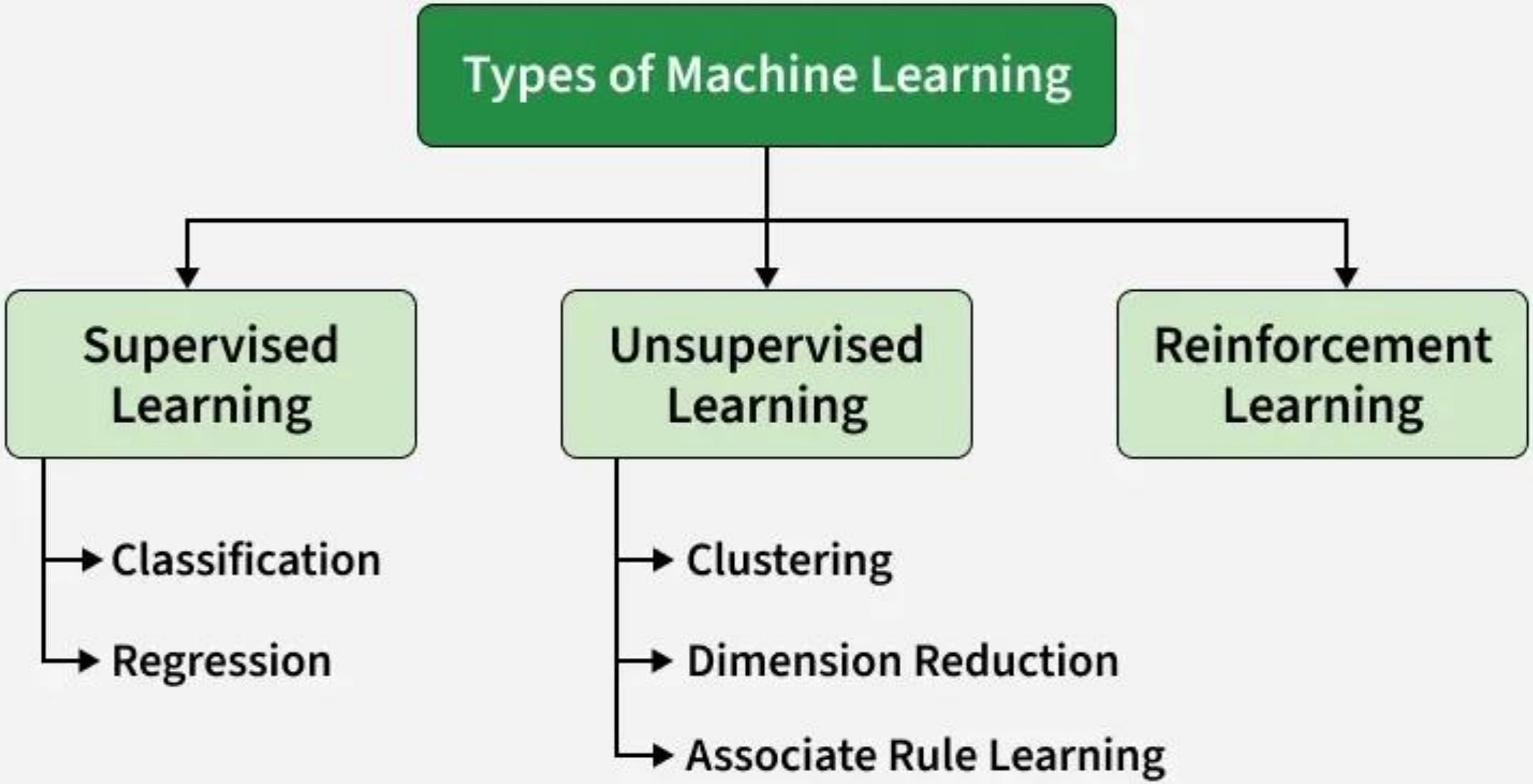
Regression



Group similar instances

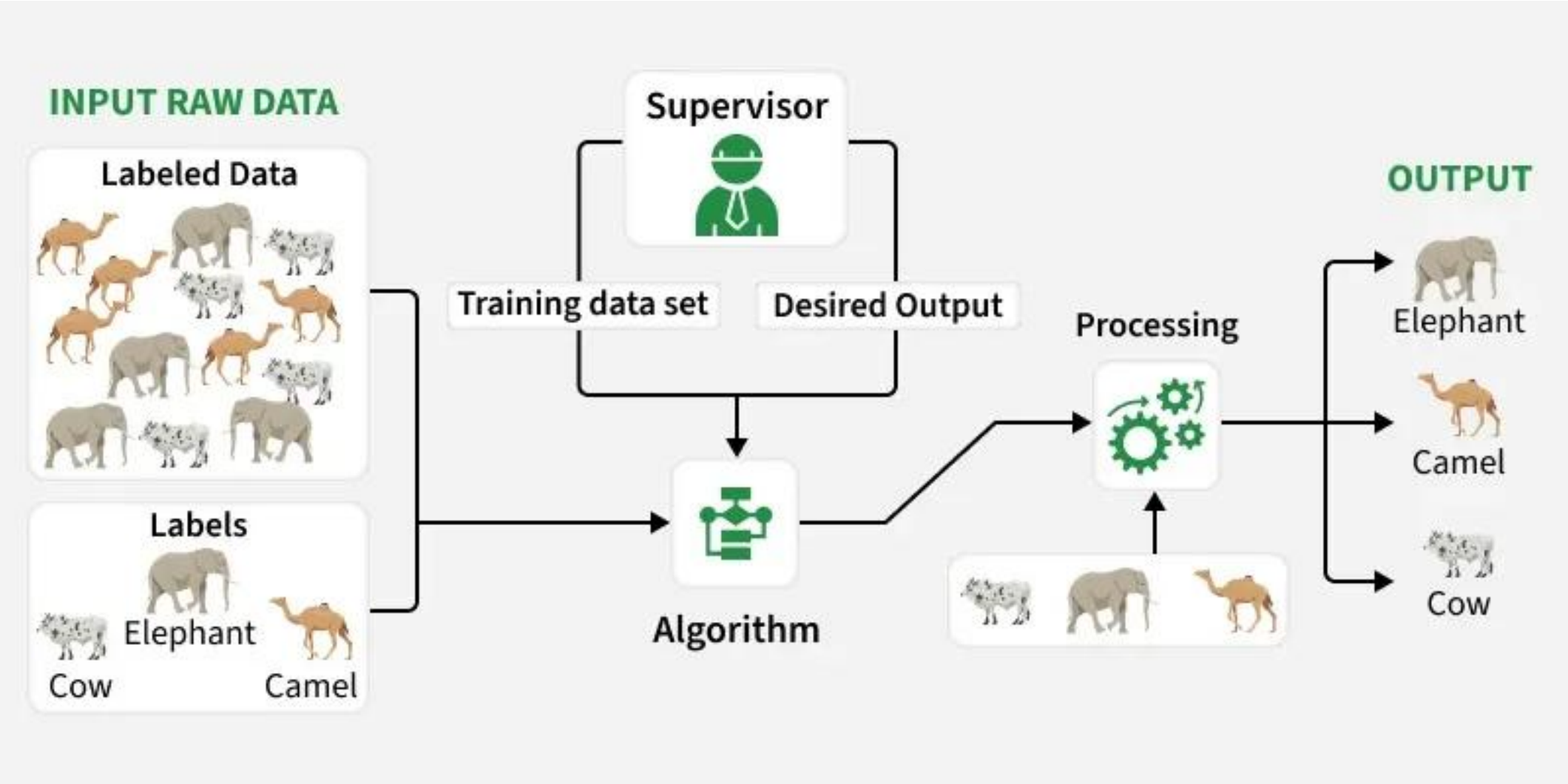
Clustering

Tasks of machine learning and data mining

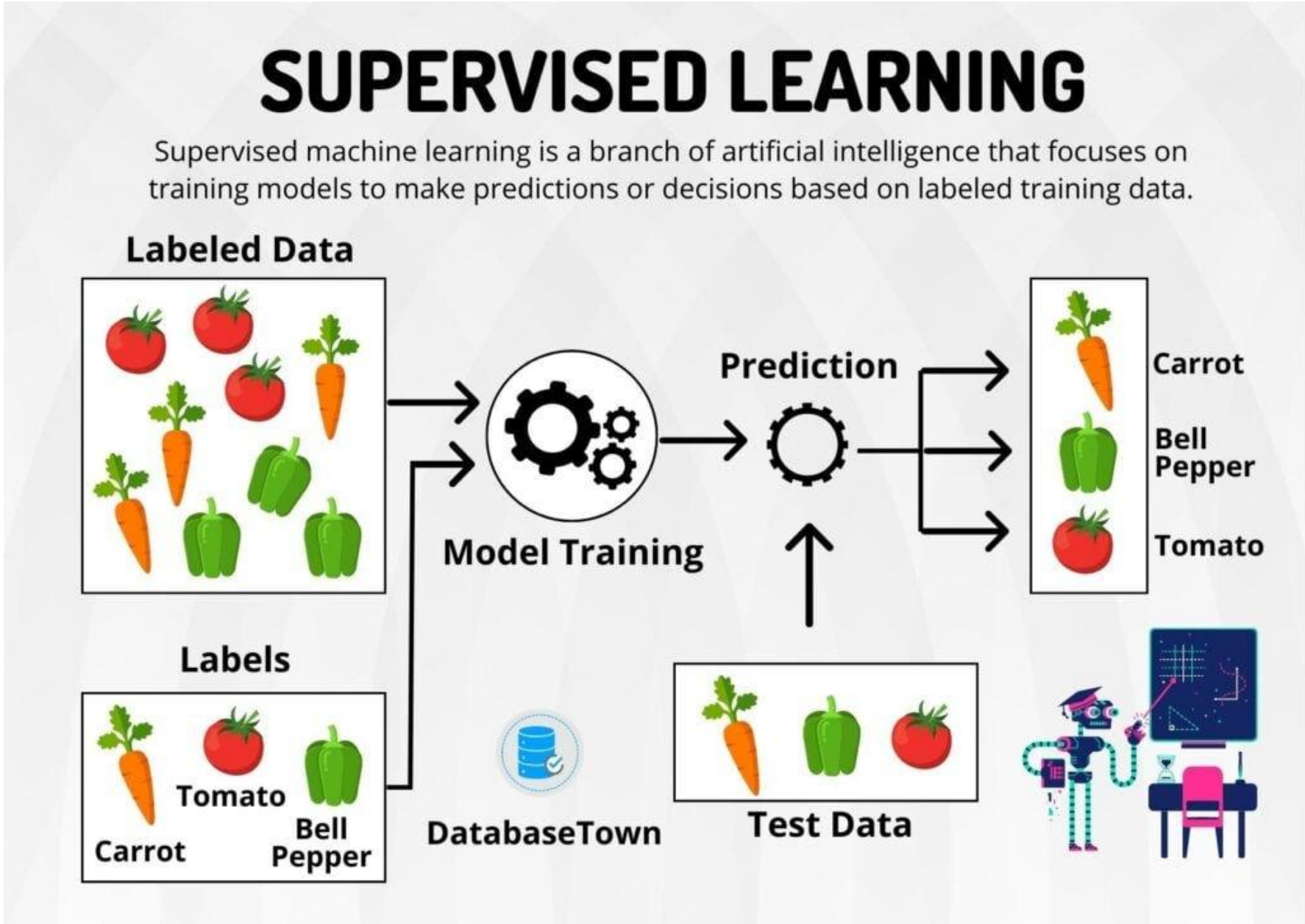


https://www.geeksforgeeks.org/machine-learning/types-of-machine-learning/

Supervised Learning

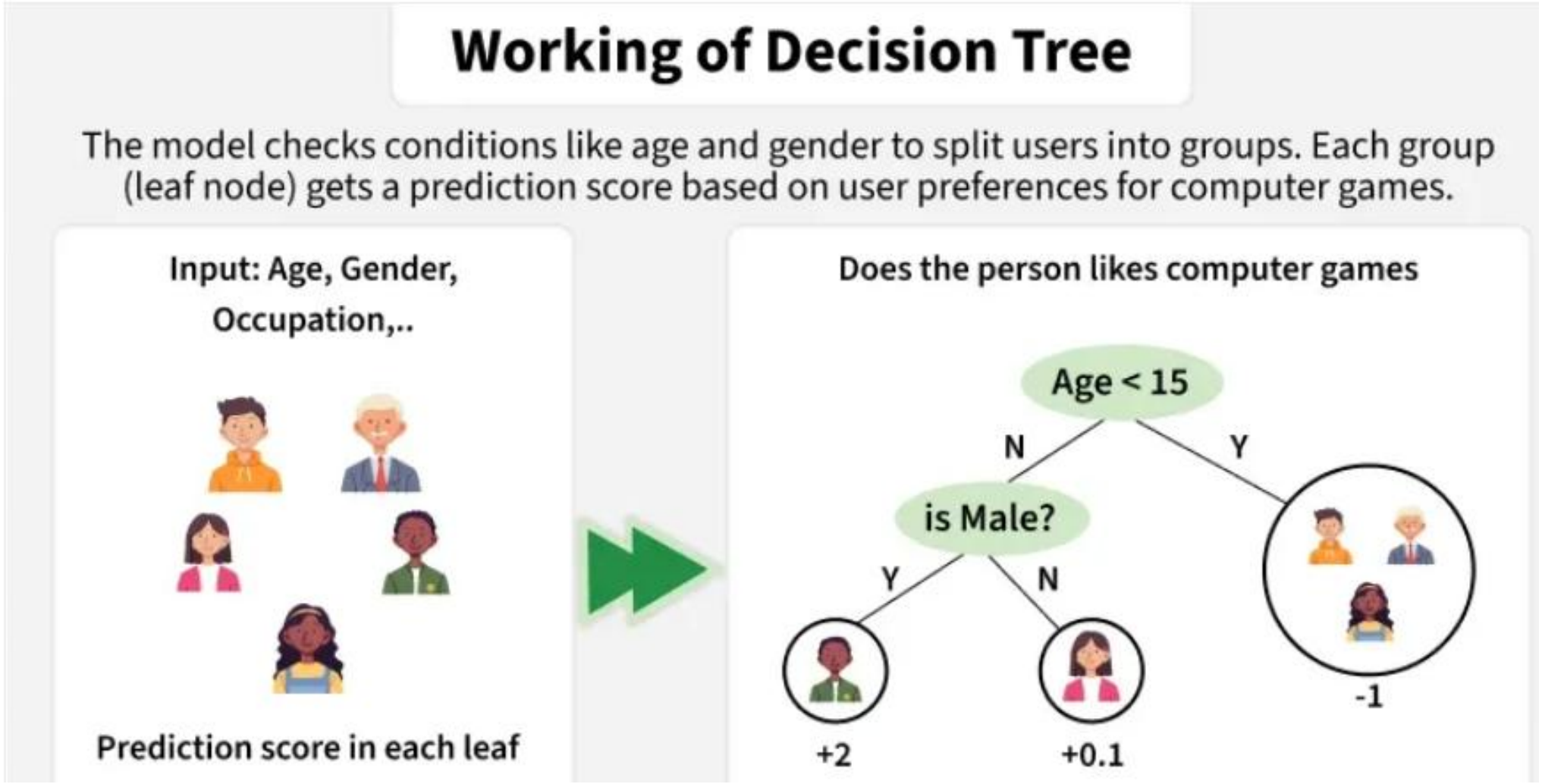


https://www.geeksforgeeks.org/machine-learning/types-of-machine-learning/



https://medium.com/@pw425/introduction-to-supervised-learning-96a6c5a98733

Supervised Learning – A Decision Tree Classifier



https://www.geeksforgeeks.org/machine-learning/decision-tree-introduction-example/

Supervised Learning Algorithms

Regression

- Linear
- Polynomial
- Ridge Regression
- Lasso Regression

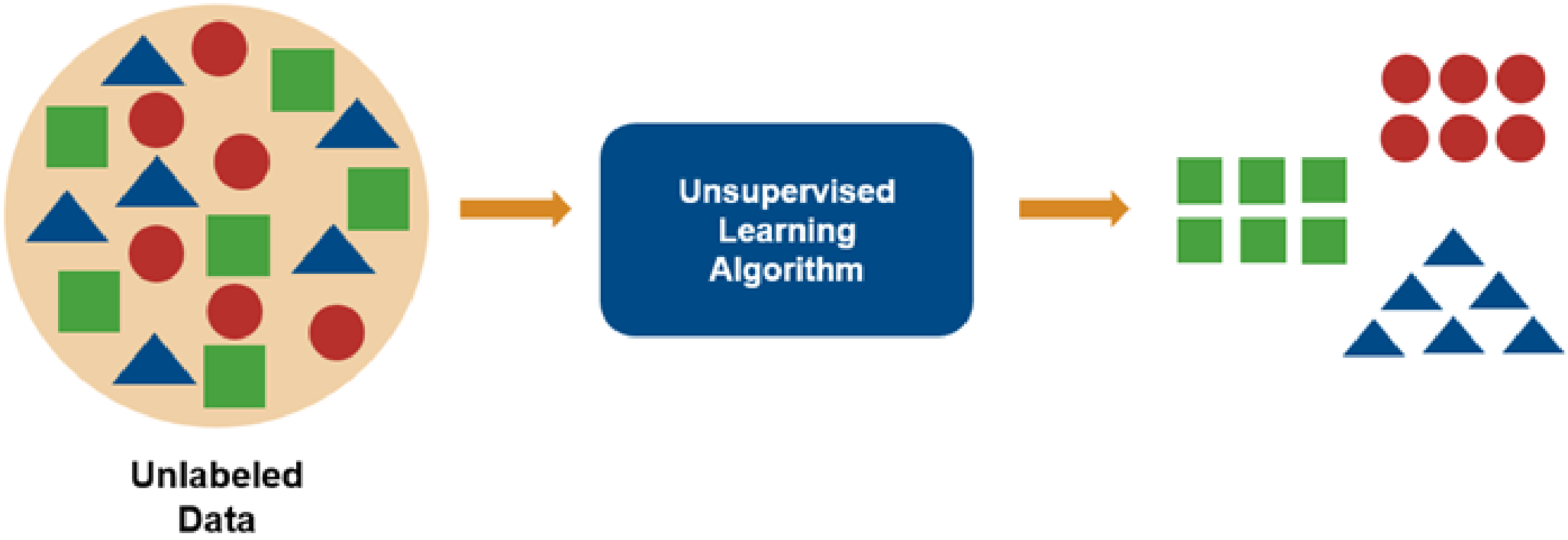
Classification

- Logistic Regression
- Linear Discriminant Analysis
- Naïve Bayes

Regression & Classification

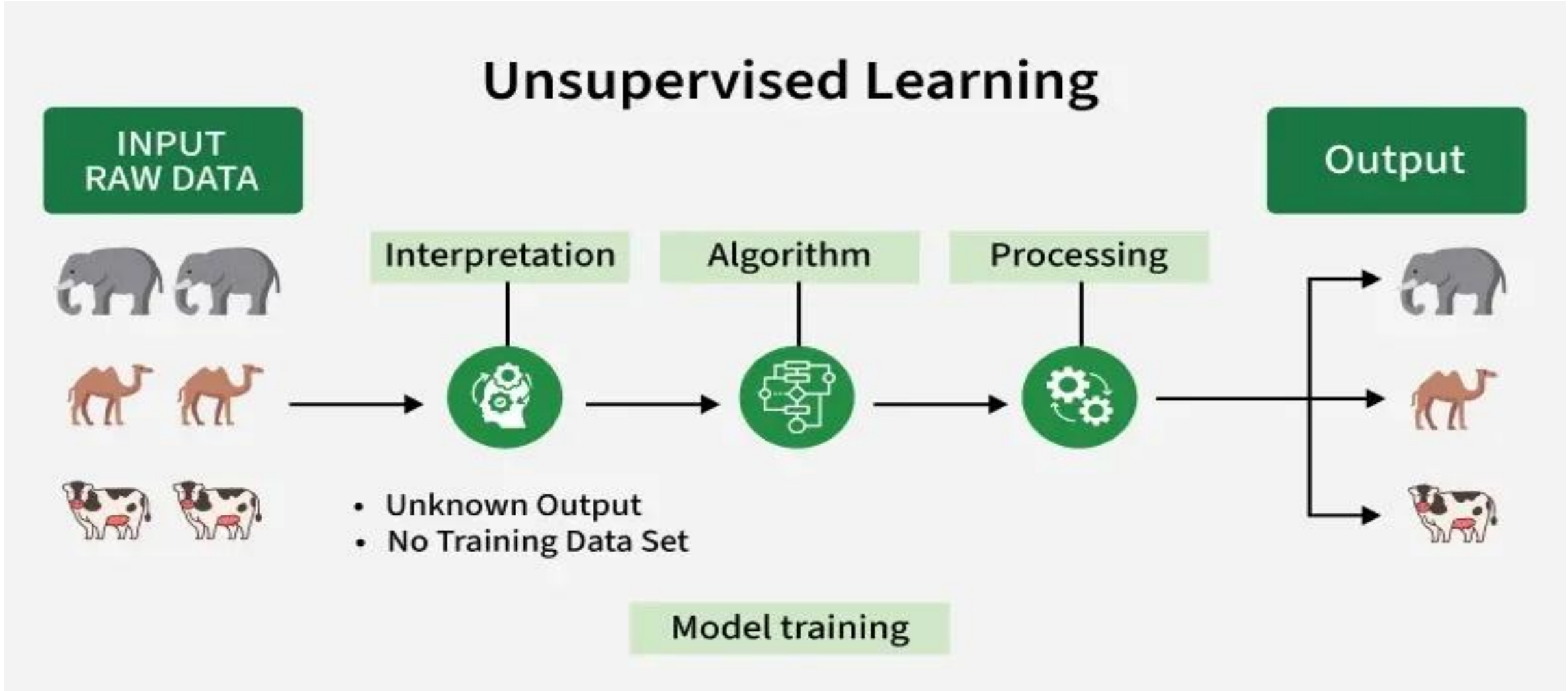
- Decision Trees
- K- Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Unsupervised Learning



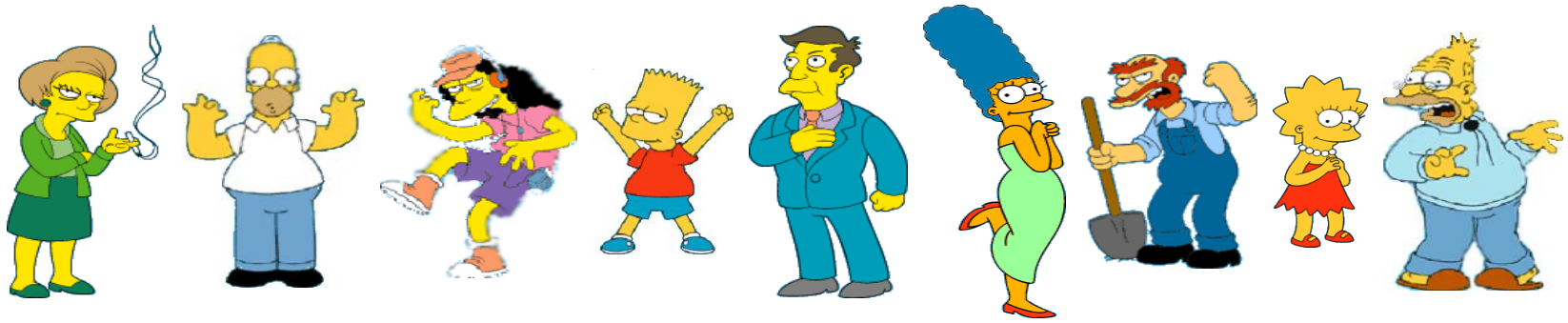
Organizing unlabeled data into groups using unsupervised learning.

https://www.mathworks.com/discovery/unsupervised-learning.html

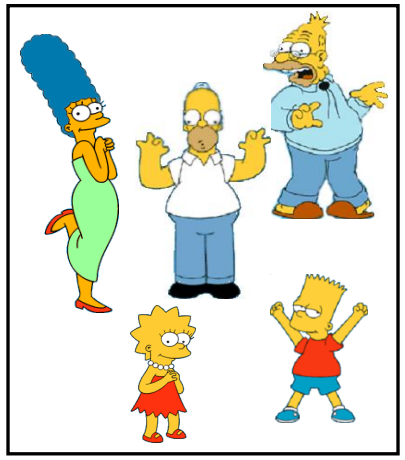


https://www.geeksforgeeks.org/machine-learning/unsupervised-learning/

Unsupervised Learning



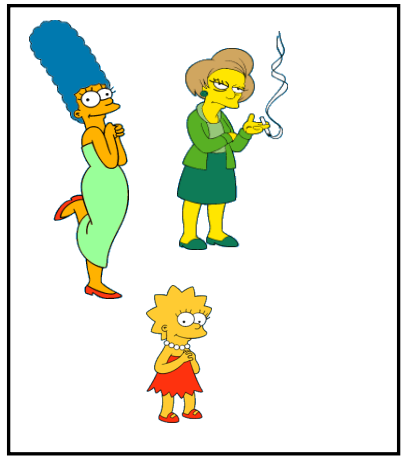
Clustering



Simpson family



School Employees



Female



Male

Unsupervised Learning

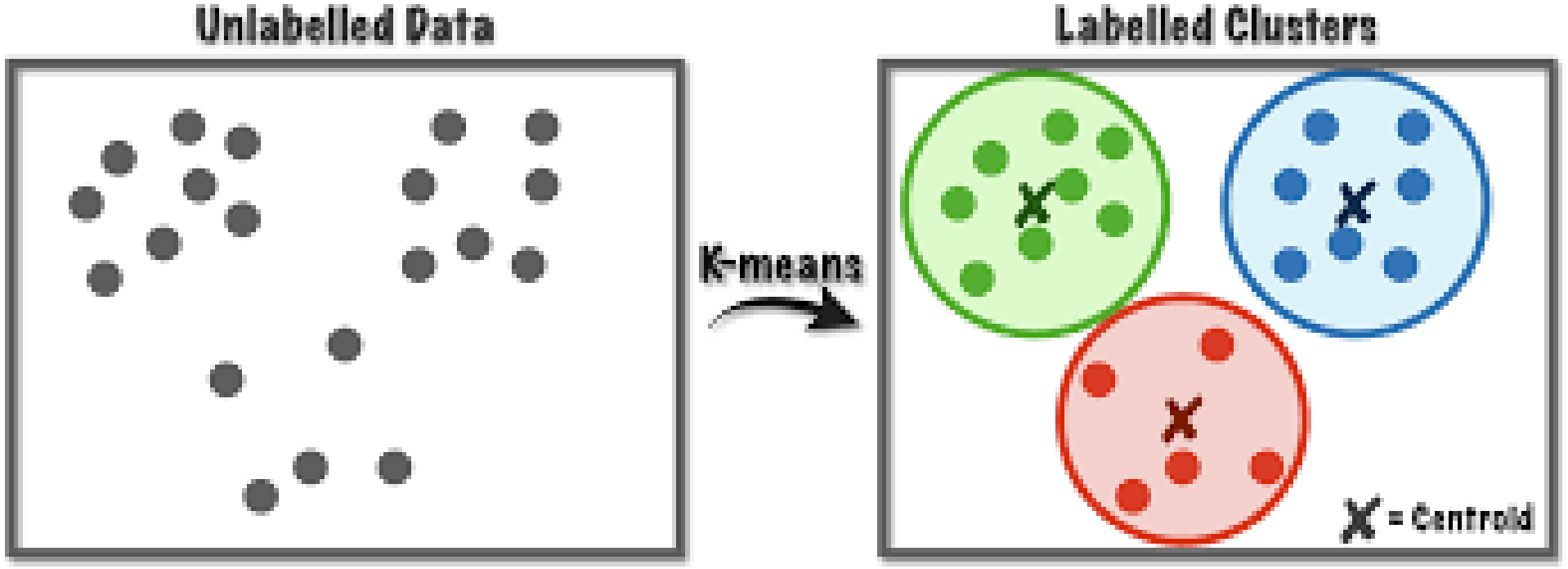
Clustering - K-Means



https://bookdown.org/tpinto_home/Unsupervised-learning/k-means-clustering.html

Unsupervised Learning

Clustering – K-Means Algorithms



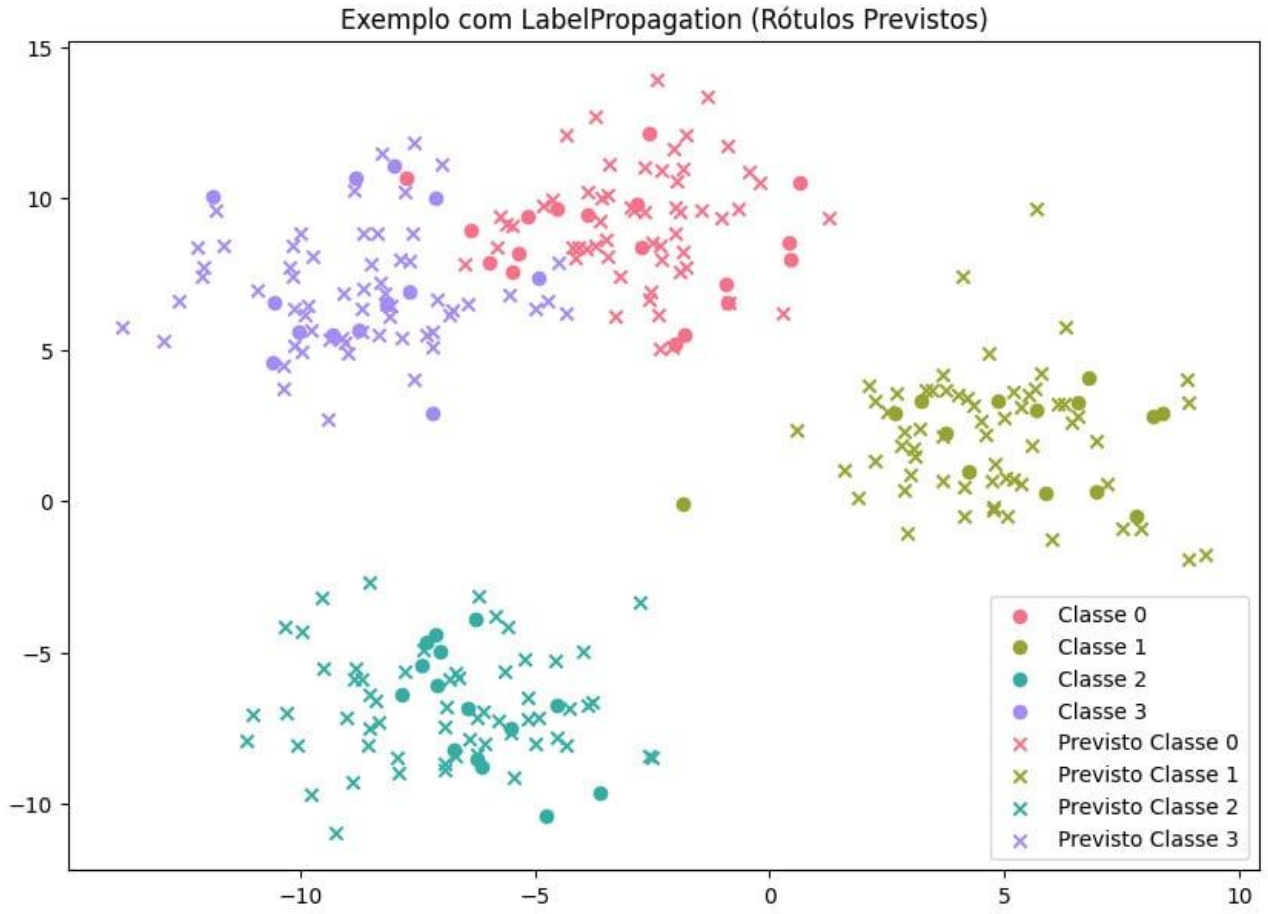
https://towardsdatascience.com/k-means-a-complete-introduction-1702af9cd8c

Semi-Supervised Learning



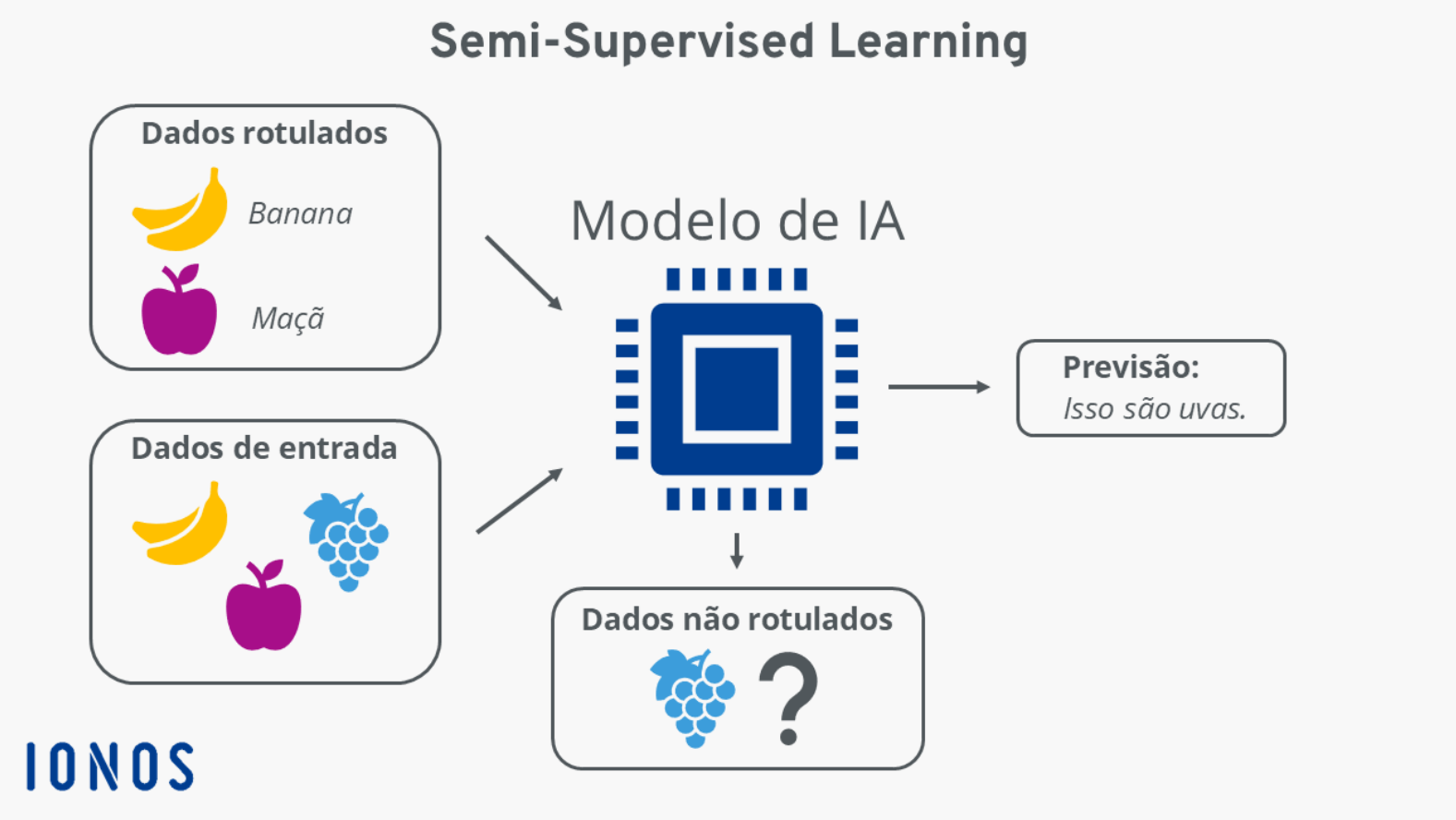
https://www.alura.com.br/artigos/machine-learning-aprendizado-semi-supervisionado?srsltid=AfmBOor5wPdKtKsQLmMciD4MmhzkqQ10LrUXmO2hzu2BwybsQEpQmgFg

Semi-Supervised Learning



https://www.alura.com.br/artigos/machine-learning-aprendizado-semi-supervisionado?srsltid=AfmBOor5wPdKtKsQLmMciD4MmhzkqQ10LrUXmO2hzu2BwybsQEpQmgFg

Semi-Supervised Learning

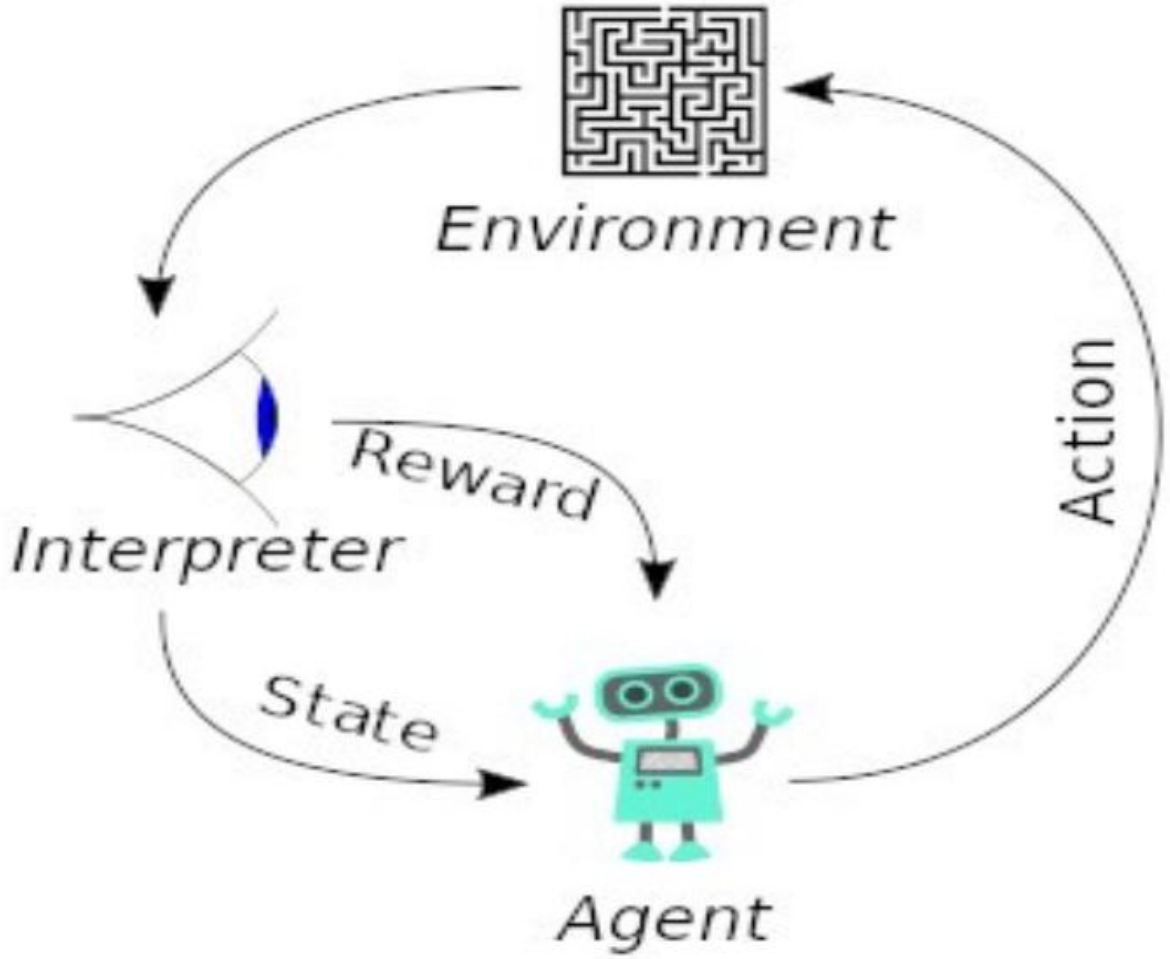


https://www.ionos.com/pt-br/digitalguide/marketing-online/marketing-em-buscadores/semi-supervised-learning/

Reinforcement Learning

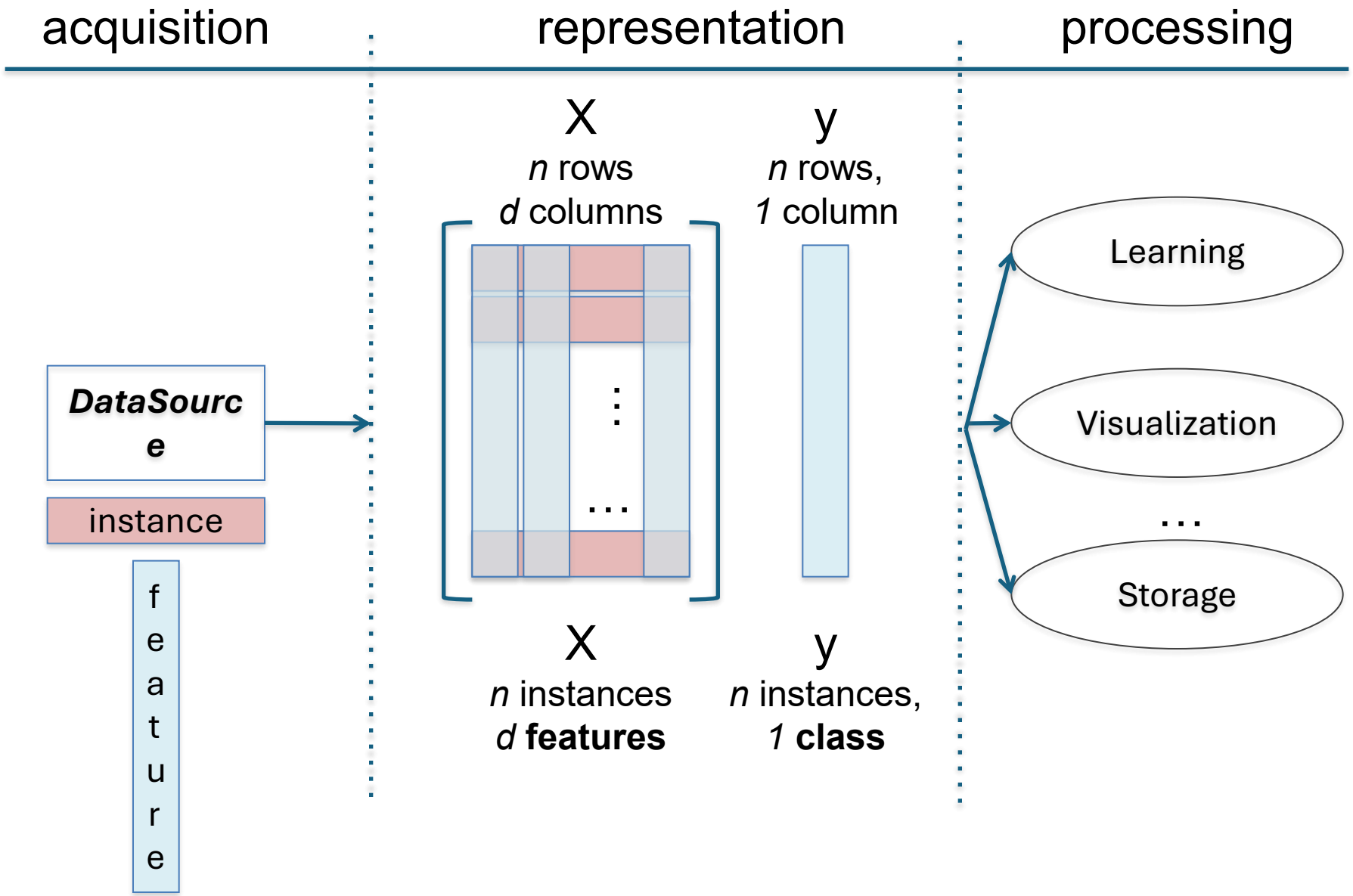
- In Reinforcement Learning (RL), agents are trained using a reward and punishment mechanism
- The agent is rewarded for correct moves and punished for incorrect ones
- Thus, the agent attempts to minimize incorrect moves and maximize correct ones
- The agent interacts with the environment and performs specific actions

Reinforcement Learning



https://neptune.ai/blog/reinforcement-learning-applications

Data and Data Mining Tasks



Data – Some Terminology

- The data is organized into instances/examples/individuals **x**
- Each instance is a vector with *d* elements **x=[x1, x2, …, x*d*]**
- Each element of **x** is the value of a **feature** (a.k.a an **attribute**)
- Each feature (attribute) represents a given measure
- A **dataset** is composed by *n* instances, each with *d* features

Outlook	Temperature	Humidity	Windy	Play
Sunny	85	85	False	No
Sunny	80	90	True	No
Overcast	83	86	False	Yes
Rainy	70	96	False	Yes
Rainy	68	80	False	Yes
Rainy	65	70	True	No

Supervised Learning

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes

- Each instance may also be described by a corresponding **class** label
  - or, in other words, we have prior-knowledge (an “oracle”) that tells us,
  - ... of **what the output values** for our samples should be
- So (in this sense) **supervised** learning is done using a **ground-truth**
- A **class** label may have a
  - binary domain (binary problem); e.g., {0, 1} or {-1, 1}, or {yes, no}, or ...
  - domain with more than two values (multiclass problem); e.g., 0, 1, 2, ..., M-1

Unsupervised Learning

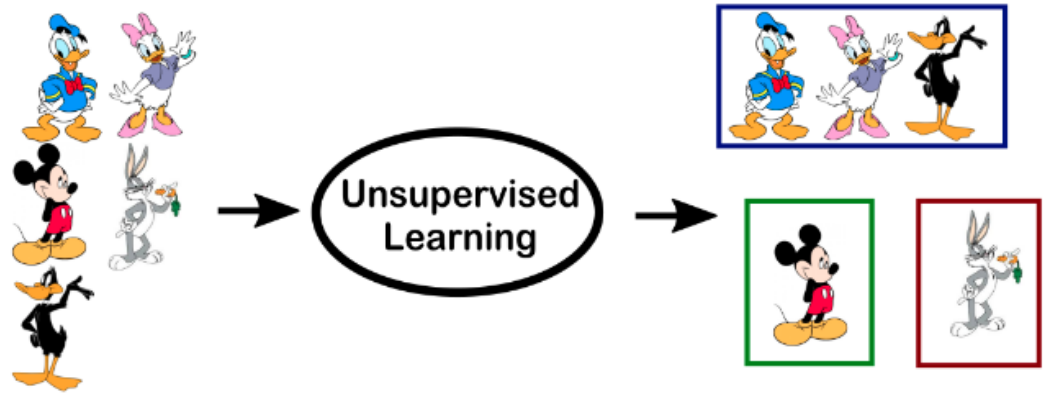
- If each instance is **NOT** described by a corresponding class label
  - we have **NO** prior-knowledge,
  - ... the **instances** themselves represent the “structure” within the data
- So (in this sense) unsupervised learning aims to **extract inherent structure** of data **without using explicitly provided labels**
- … therefore, unsupervised learning is useful in **exploratory analysis** because it can automatically **identify structure** in data
  - e.g., if an analyst were trying to segment consumers, unsupervised clustering methods would be a starting point
  - … where it is impossible or impractical for a human to propose trends in data, unsupervised learning can provide initial insights that can then be used to test individual hypotheses

Unsupervised Learning

search for "structural" resemblances

among

instances and group them together...

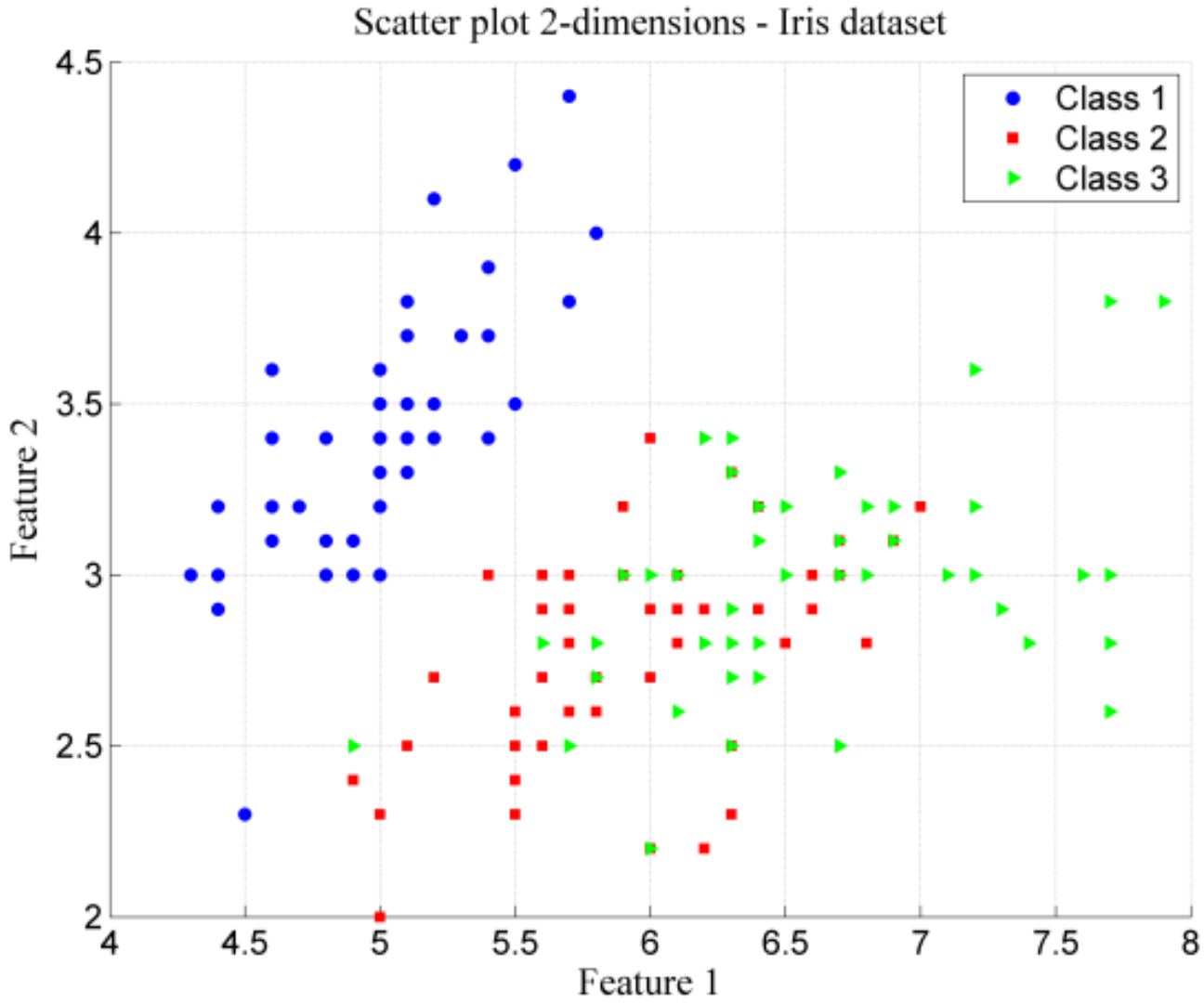


	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
...				
51	7.0	3.2	4.7	1.4
52	6.4	3.2	4.5	1.5
53	6.9	3.1	4.9	1.5
54	5.5	2.3	4.0	1.3
55	6.5	2.8	4.6	1.5
...				
101	6.3	3.3	6.0	2.5
102	5.8	2.7	5.1	1.9
103	7.1	3.0	5.9	2.1
104	6.3	2.9	5.6	1.8
105	6.5	3.0	5.8	2.2

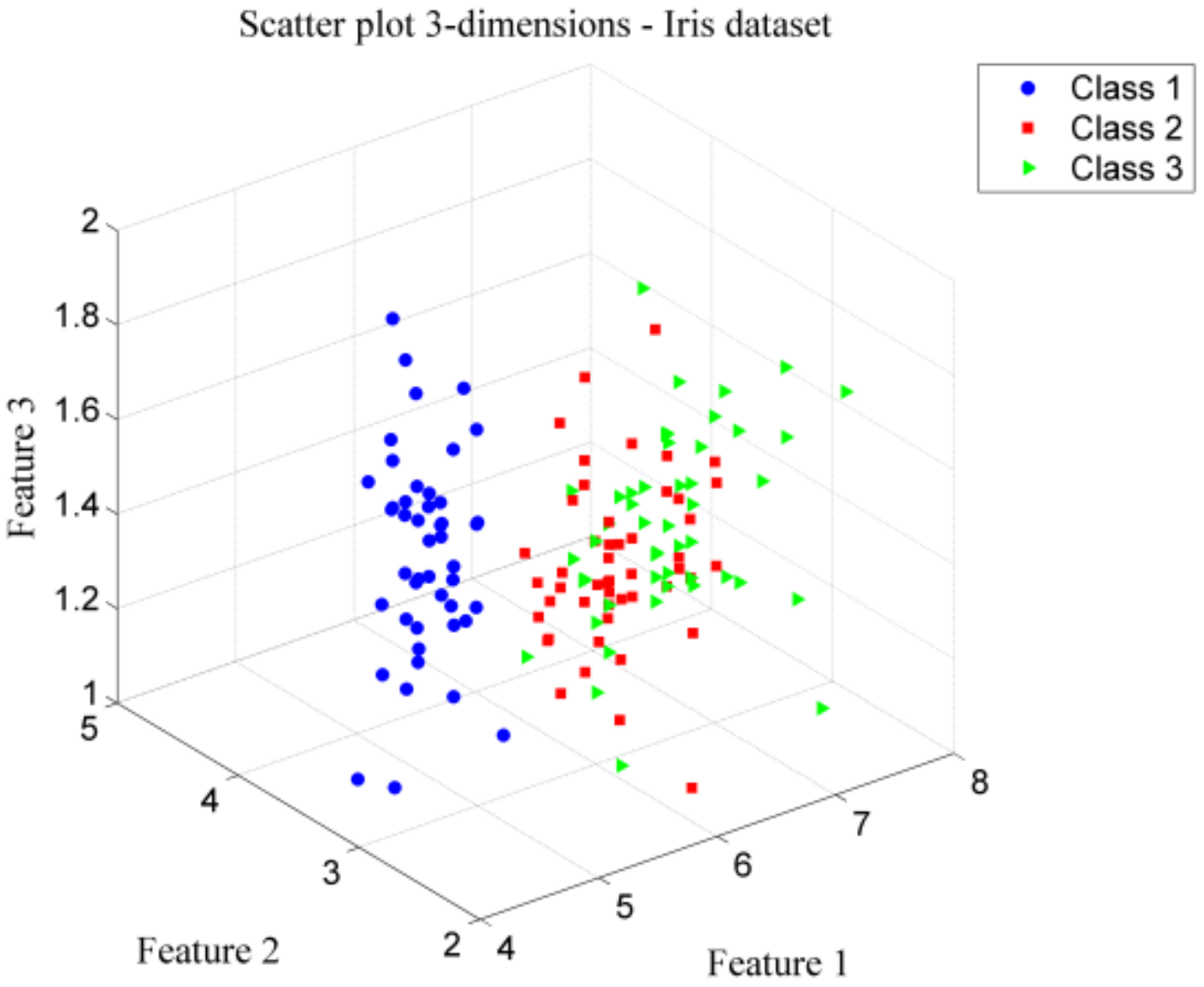
... Data – about the Features (Attributes)

- Should be in lowest number as possible
- Should be as discriminative as possible
- Should be relevant
- Should not be redundant, on the presence of others

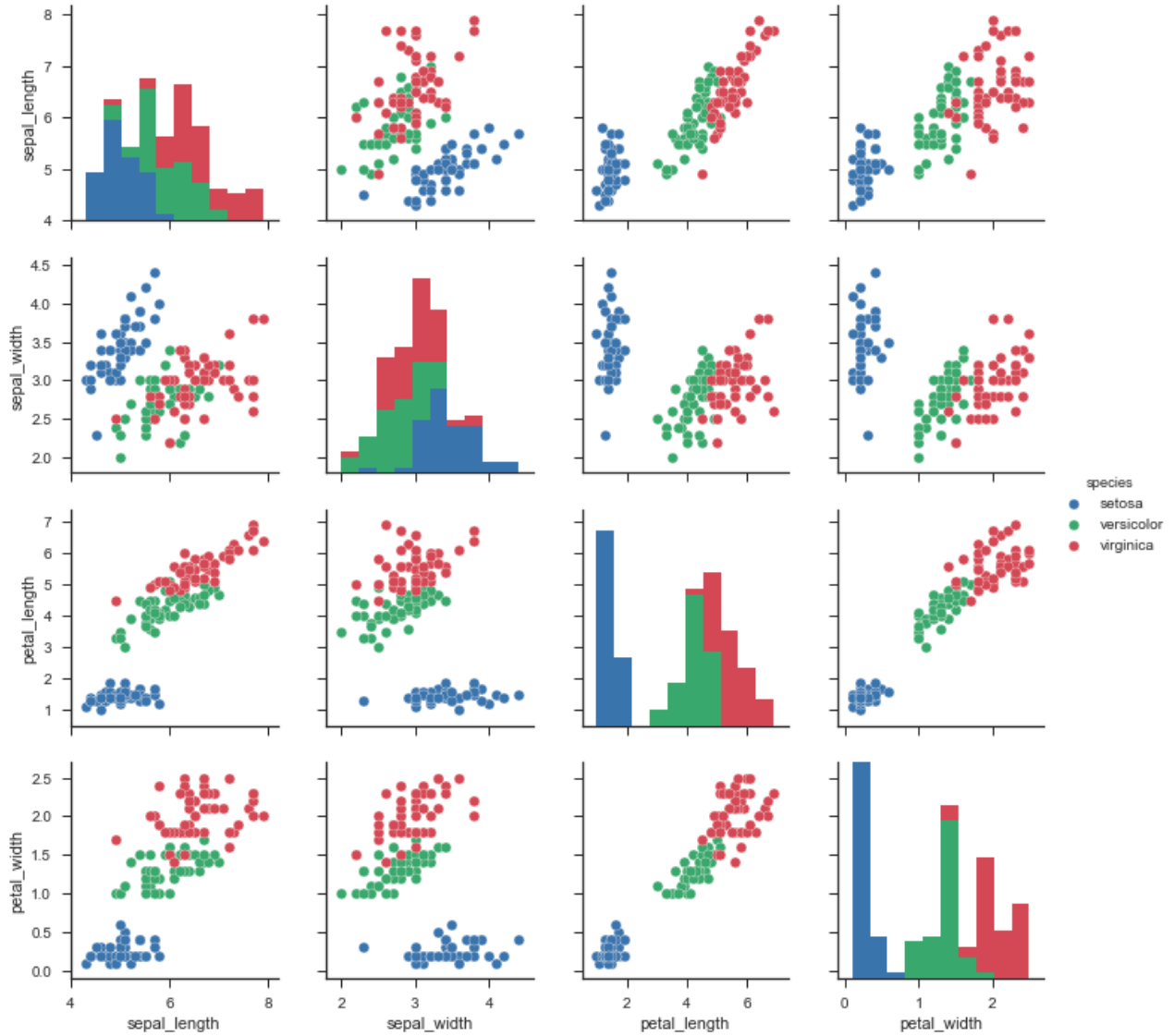
Data – about Visualization (2 features, 3 classes)



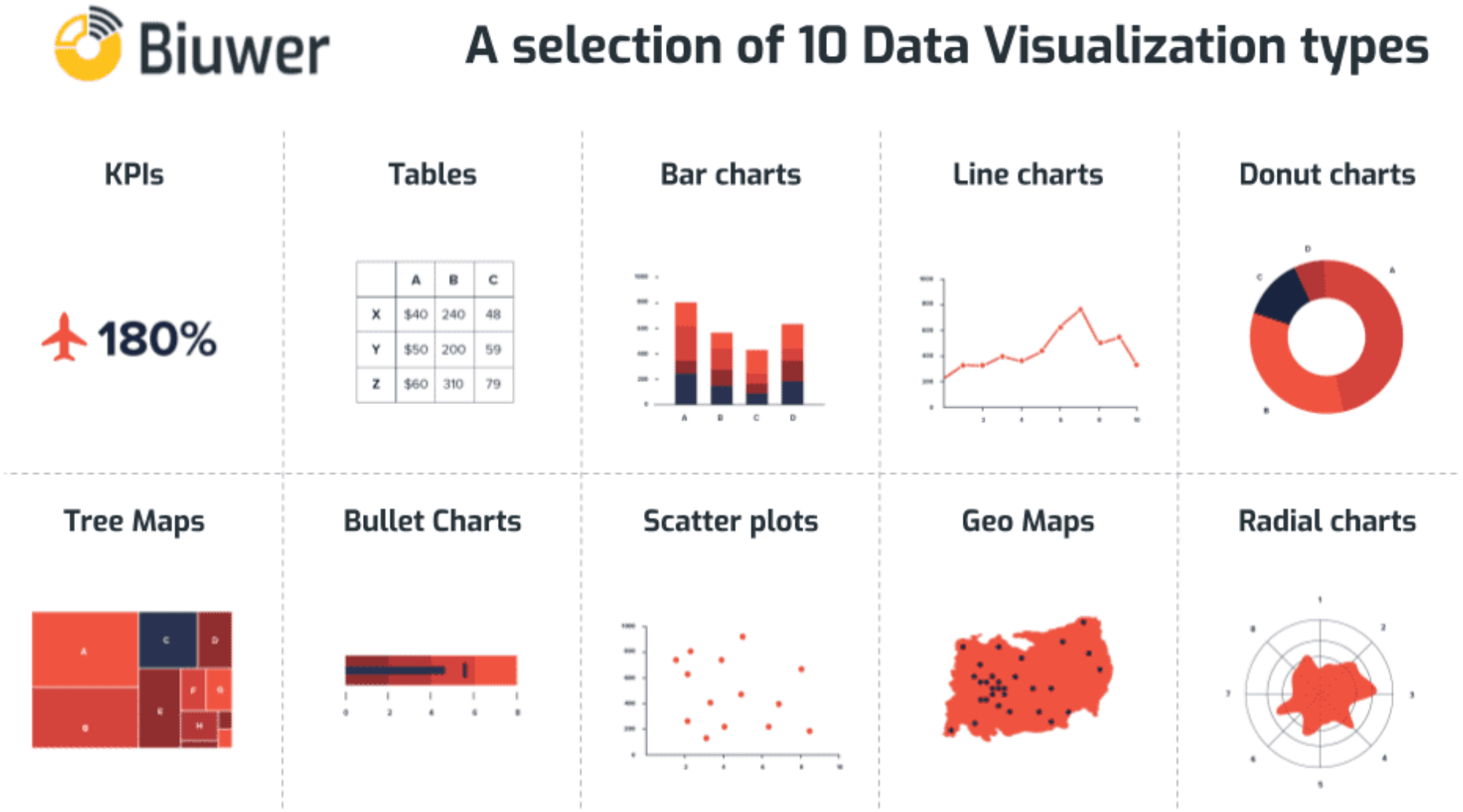
Data – about Visualization (3 features, 3 classes)



Data – about Visualization (all 2 features, 3 classes)

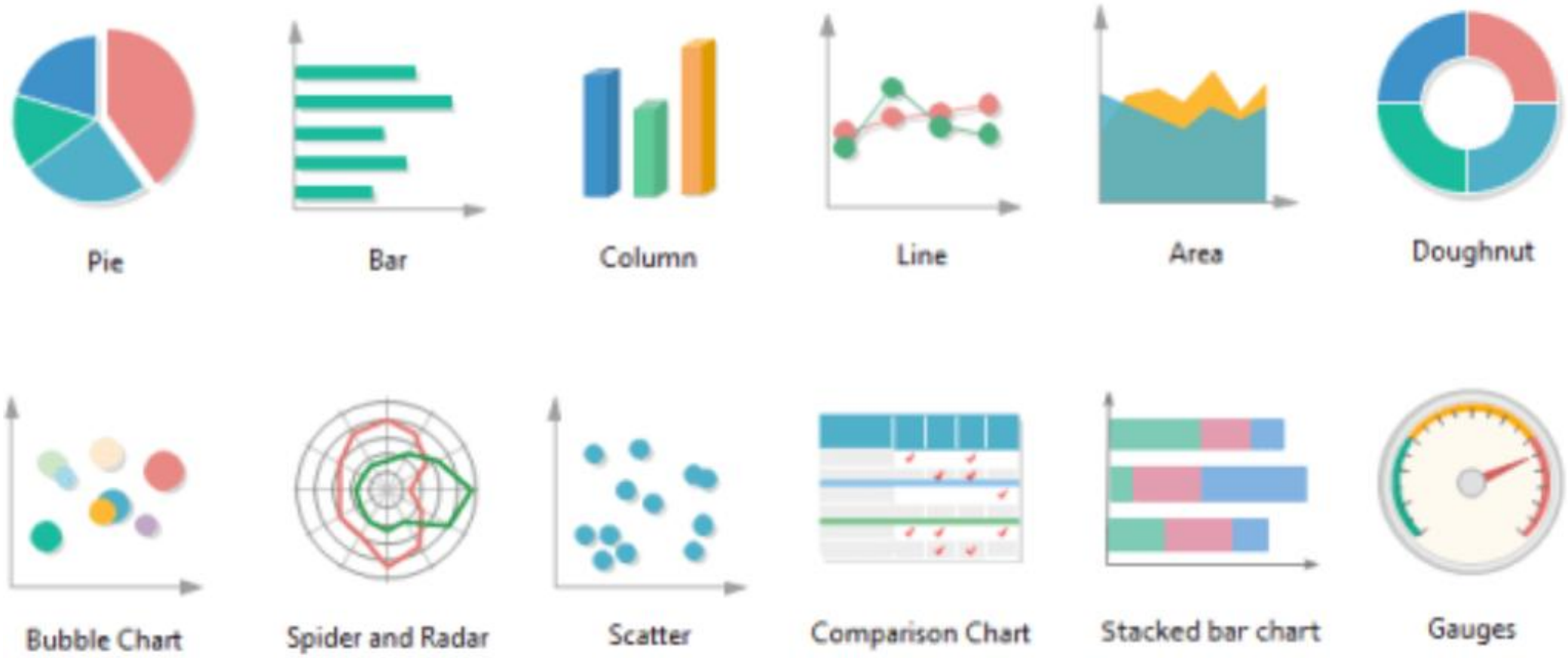


Data – Visualization



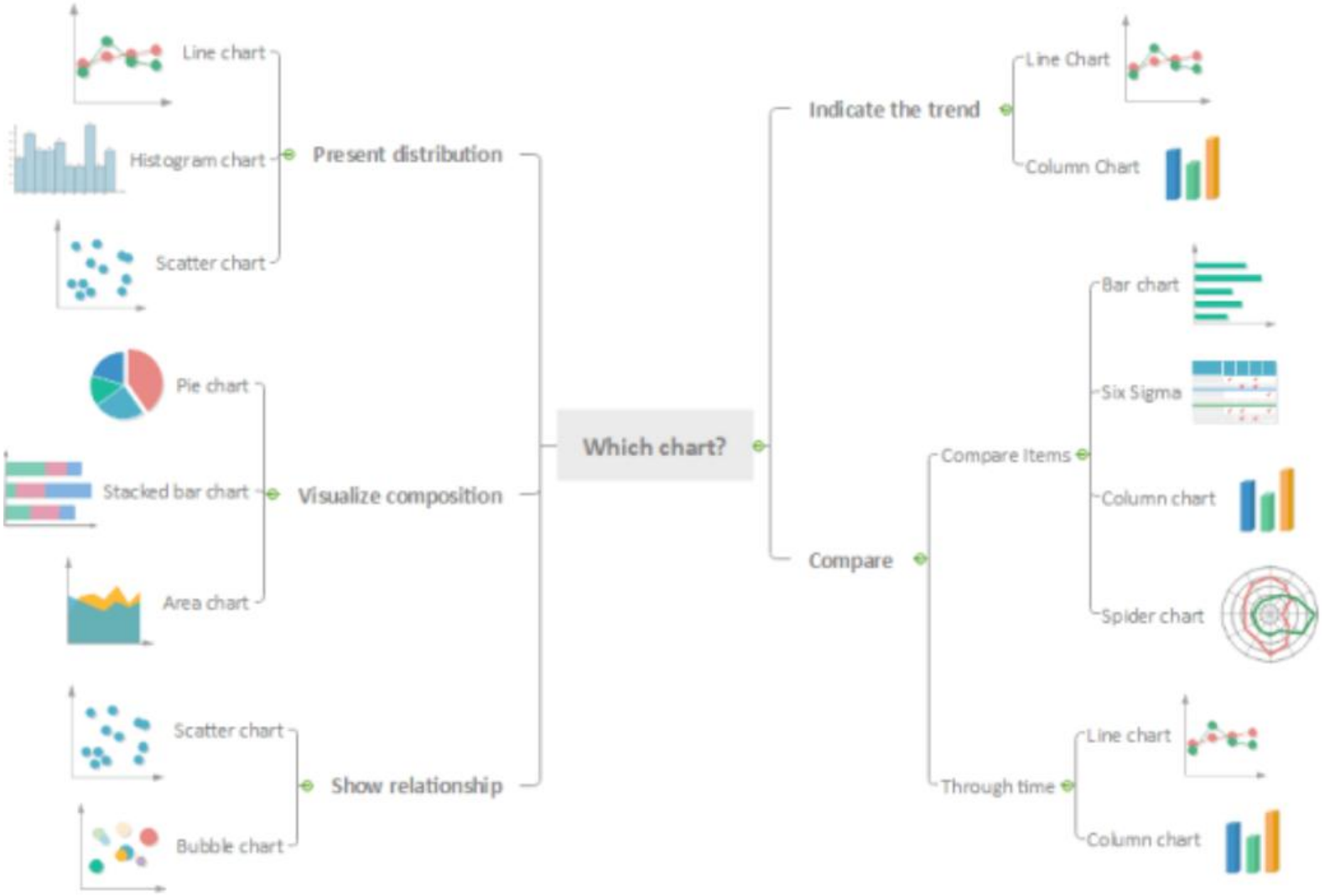
https://biuwer.com/en/blog/what-is-data-visualization-from-data-to-visualization/

Data – Visualization



https://codecrucks.com/what-is-data-visualization/
https://www.edrawsoft.com/chart/choose-right-chart.html

Data – Visualization



https://codecrucks.com/what-is-data-visualization/
https://www.edrawsoft.com/chart/choose-right-chart.html

Data Representation (ARFF)

- ARFF (Attribute-Relation File Format) file is an **ASCII text** file that describes a list of **instances** sharing a set of **attributes**.
- ARFF files have two distinct sections:
  - the first section is the **Header** (*meta-data*)
  - the second section is the **Data**.

ARFF files were developed by the Machine Learning (ML) Project at the Department of Computer Science of The University of Waikato for use with the Weka ML software.

`https://www.cs.waikato.ac.nz/ml/weka/arff.html`

`https://waikato.github.io/weka-wiki/formats_and_processing/arff_stable/`

Data Representation (ARFF – header example)

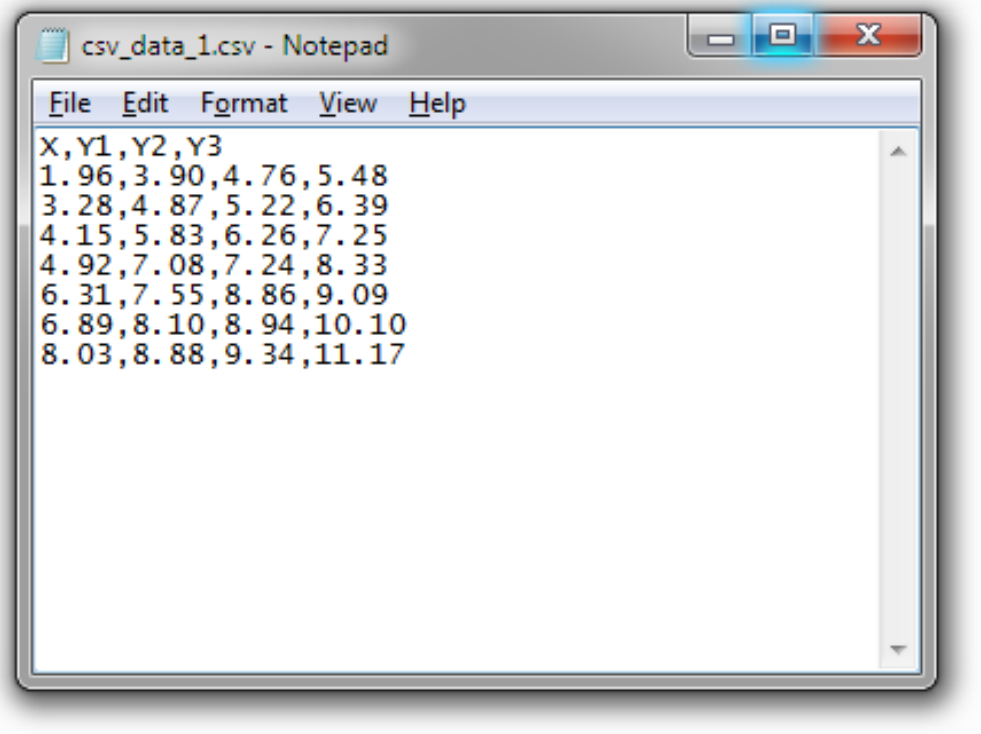
```
% 1. Title: Iris Plants Database
%
% 2. Sources:
% (a) Creator: R.A. Fisher
% (b) Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)
% (c) Date: July, 1988
%
@RELATION iris

@ATTRIBUTE sepallength NUMERIC
@ATTRIBUTE sepalwidth NUMERIC
@ATTRIBUTE petallength NUMERIC
@ATTRIBUTE petalwidth NUMERIC
@ATTRIBUTE class {Iris-setosa,Iris-versicolor,Iris-virginica}
```

Data Representation (ARFF – data example)

```
@DATA
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
4.7,3.2,1.3,0.2,Iris-setosa
4.6,3.1,1.5,0.2,Iris-setosa
5.0,3.6,1.4,0.2,Iris-setosa
5.4,3.9,1.7,0.4,Iris-setosa
4.6,3.4,1.4,0.3,Iris-setosa
5.0,3.4,1.5,0.2,Iris-setosa
4.4,2.9,1.4,0.2,Iris-setosa
4.9,3.1,1.5,0.1,Iris-setosa
...
```

Data Representation
Comma Separated Value (CSV) file



Sales_Data.csv — Edited

```
2024-12-12;Central;Douglas;John;Television;67;$1 198,00;$80 266,00
2024-12-29;East;Douglas;Karen;Video Games;74;$58,50;$4 329,00
2024-01-15;Central;Timothy;David;Home Theater;46;$500,00;$23 000,00
2024-02-01;Central;Douglas;John;Home Theater;87;$500,00;$43 500,00
2024-02-18;East;Martha;Alexander;Home Theater;4;$500,00;$2 000,00
2024-03-07;West;Timothy;Stephen;Home Theater;7;$500,00;$3 500,00
2024-03-24;Central;Hermann;Luis;Video Games;50;$58,50;$2 925,00
2024-04-10;Central;Martha;Steven;Television;66;$1 198,00;$79 068,00
2024-04-27;East;Martha;Diana;Cell Phone;96;$225,00;$21 600,00
2024-05-14;Central;Timothy;David;Television;53;$1 198,00;$63 494,00
2024-05-31;Central;Timothy;David;Home Theater;80;$500,00;$40 000,00
2024-06-17;Central;Hermann;Shelli;Desk;5;$125,00;$625,00
2024-07-04;East;Martha;Alexander;Video Games;62;$58,50;$3 627,00
2024-07-21;Central;Hermann;Sigal;Video Games;55;$58,50;$3 217,50
2024-08-07;Central;Hermann;Shelli;Video Games;42;$58,50;$2 457,00
2024-08-24;West;Timothy;Stephen;Desk;3;$125,00;$375,00
2024-09-10;Central;Timothy;David;Television;7;$1 198,00;$8 386,00
2024-09-27;West;Timothy;Stephen;Cell Phone;76;$225,00;$17 100,00
2024-10-14;West;Douglas;Michael;Home Theater;57;$500,00;$28 500,00
2024-10-31;Central;Martha;Steven;Television;14;$1 198,00;$16 772,00
2024-11-17;Central;Hermann;Luis;Home Theater;11;$500,00;$5 500,00
2024-12-04;Central;Hermann;Luis;Home Theater;94;$500,00;$47 000,00
2024-12-21;Central;Martha;Steven;Home Theater;28;$500,00;$14 000,00
```

Data Representation (Orange-DM)

- Orange-DM native data format is a tab-delimited text file with three header (*meta-data*) rows:
  - first row lists **attribute** names
  - second row defines their **domain** (continuous – c, discrete – d and string – s)
  - third row is an optional type (class, meta, or ignore)
- Orange-DM also supports a condensed single-line header format
  - feature names prefixed by an optional “<flags>#” string

Orange-DM (Data Mining) is an open source machine learning and data visualization.

Allows to build data analysis workflows visually, with a large, diverse toolbox.

`https://orangedatamining.com/`

`https://orange3.readthedocs.io/projects/orange-data-mining-library/en/latest/reference/data.io.html/`

Data Representation (Orange-DM example)

sepal length	sepal width	petal length	petal width	iris
c	c	c	c	d
				class
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3.0	1.4	0.2	Iris-setosa
4.7	3.2	1.3	0.2	Iris-setosa
4.6	3.1	1.5	0.2	Iris-setosa

age	prescription	astigmatic	tear_rate	lenses
discrete	discrete	discrete	discrete	discrete
				class
young	myope	no	reduced	none
young	myope	no	normal	soft
young	myope	yes	reduced	none
young	myope	yes	normal	hard
young	hypermetrope	no	reduced	none

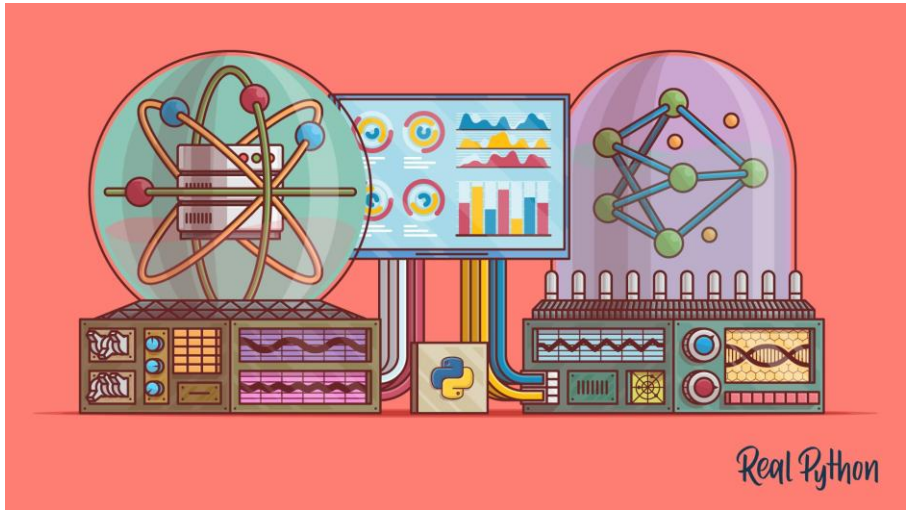
Software
Orange Data Mining



https://orangedatamining.com/

https://www.youtube.com/watch?v=HXjnDIgGDuI&list=PLmNPvQr9Tf-ZSDLwOzxpvY-HrE0yv-8Fy

Python





https://realpython.com/learning-paths/data-science-python-core-skills/

https://www.linkedin.com/pulse/getting-started-python-data-science-five-ruthlessly-effective-ken-vu/



Python - Introduction

https://www.w3schools.com/python/python_intro.asp

https://www.geeksforgeeks.org/data-science/data-science-with-python-tutorial/

https://aeturrell.github.io/python4DS/welcome.html

About Data Science

https://www.w3schools.com/datascience/default.asp

https://www.w3schools.com/datascience/ds_introduction.asp

https://www.w3schools.com/datascience/ds_data.asp

http://w3schools.com/datascience/ds_python.asp

https://www.geeksforgeeks.org/data-science/introduction-to-data-science/

https://towardsdatascience.com/your-first-90-days-as-a-data-scientist/